A thesis submitted in partial fulfillment of the requirements for the degree Master of Science

Determination of the Discovery Potential of Leptoquarks Decaying into a Tau-lepton and a Top-quark Using the ATLAS Detector at CERN

Ishaan Utkarsh
born in Jharkhand, India

October 7, 2025

Fakultät Physik (Technische Universität Dortmund)

Dipartimento di Fisica e Astronomia (Università di Bologna)

École Universitaire de Physique et d’ Ingénierie (Université Clermont Auvergne)

This thesis summarizes the scientific work carried out at the Chair of Experimental Physics of Technische Universität Dortmund.
Supervisor: Assoc. Prof. Dr. André Sopczak
First reviewer: Assoc. Prof. Dr. André Sopczak
Second reviewer: Dr. Carsten Burgard

Date of submission: September 2025

To Mummy, Papa and Bhaiya

Acknowledgement

I would like to express my deepest gratitude to my supervisor, Assoc. Prof. Dr. André Sopczak, for his continuous guidance and support throughout this project. His valuable insights and explanations were instrumental in helping me navigate and understand the intricacies of the analysis. I am also sincerely thankful to Stergios Kazakos and Gabriel Oliveira Correa from the Tau+X Group at CERN for their constant guidance and for patiently assisting me with the complexities of the ATLAS analysis frameworks.

My sincere thanks go to Dr. Babar Ali for sharing his expertise and providing important insights into the fake-lepton analysis. I am also grateful to Viktoriia Lysenko for guiding me through the process of requesting Monte Carlo samples and for her help in understanding the details of the CERN JIRA portal. I would like to acknowledge Yanlin Liu for taking responsibility for managing the production of the signal Monte Carlo samples used in this analysis.

I am especially thankful to my colleague and friend, Adwait Meppurath, for the countless discussions we shared about the analysis, as well as for the much-needed ping-pong matches that brightened the long and sometimes monotonous days in Prague. I am equally indebted to all my friends who supported and encouraged me, helping me stay balanced and motivated throughout the two years of this Master’s programme.

Finally, I would like to express my heartfelt gratitude to my family — my mother, father, and brother — for their unwavering belief in me and their unconditional support. Without their encouragement and love, I would not have been able to reach this stage.

Abstract

Leptoquarks are BSM particles that couple leptons and quarks at tree level. They naturally arise in GUT unification models comprising of higher symmetries like $SU(5)$ or $SO(10)$ that unify quarks and leptons. Recently, interest in leptoquarks has increased after the observations of B-physics anomalies. This thesis focuses on the development of an experimental pipeline with high sensitivity towards the novel asymmetric pair production of scalar leptoquarks in the $2\ell SS1\tau$ channel in high energy proton-proton collisions at $\sqrt{s} = 13\,\text{TeV}$. Run 2 data with a luminosity of $140\,fb^{-1}$ was used to evaluate data-MC agreement. Three different sets of signal MC samples were produced for modeling the signal, in total comprising of 28 mass points—11 mass points for same mass hypothesis and Yukawa coupling $y = 0.5$, 11 mass points for same mass hypothesis and Yukawa coupling $y = 1.0$ and 6 mass points for different mass hypothesis and Yukawa coupling $y = 1.0$. The same mass samples had the leptoquark mass range from $1500 - 2500\,\text{GeV}$ in steps of $100\,\text{GeV}$ and the different mass samples had mass points of $1500\,\&\,2500$, $1700\,\&\,2300$, $1900\,\&\,2100$, $2100\,\&\,1900$, $2300\,\&\,1700$ and $2500\,\&\,1500\,\text{GeV}$. Pre-Produced MC samples were used to model background. An extensive object selection and event selection was performed and the events were re-classified into prompt and fake leptons. Four orthogonal control regions were defined to model the agreement between background and real data and best fit values of norm factors for major fake lepton contributions were obtained using the template fit method. The obtained values were similar across all the signal models, ensuring stability of background modeling. A TabNet model was trained in the signal region to calculate signal probability with signal defined as all the signal smaples and background defined as all the SM processes. The classifier showed an impressive performance in separating background and signal with an $AUC = 0.9995$. Expected 95% CL upper limits were obtained by performing a profile likelihood fit in the signal region on the effective mass and the signal probability. The results obtained using signal probability were observed to be slightly better for lower masses for the same mass hypothesis and better by a factor of about 0.6 for different mass hypotheses. The analysis resulted in an expected exclusion of leptoquark masses less than about $2000\,\text{GeV}$ for $y = 2.0$ and an expected exclusion of all the masses in the range $1500 - 2500\,\text{GeV}$ for different mass hypothesis and $y = 2.0$. It was also established by the analysis that the kinematics of the signal samples is independent of the value of the Yukawa coupling.

Keywords: Leptoquarks, Grand Unified Theory, Asymmetric Pair Production, CERN, LHC, ATLAS, Machine Learning

Contents

List of Figures

List of Tables

1 Introduction

The Standard Model of Particle Physics [1–3] stands as the most comprehensive theoretical framework for elucidating the fundamental interactions within the universe. In 2012, the ATLAS and CMS collaborations at CERN made a landmark discovery of a Higgs-like particle, marking a pivotal moment in the validation of the Standard Model's theoretical underpinnings [4, 5]. This model has emerged as a significant achievement of contemporary physics, offering predictions that align with experimental outcomes to an extraordinary degree of precision. However, despite its remarkable successes, the Standard Model is confronted with notable limitations, as evidenced by persistent discrepancies between its theoretical predictions and various experimental observations. These inconsistencies have prompted the exploration of Beyond the Standard Model (BSM) theories, which introduce novel symmetries and additional particles to reconcile these anomalies. Among the proposed extensions to the Standard Model are leptoquarks, a class of hypothetical particles that present a promising avenue for further investigation into the fundamental forces and the structure of matter.

Leptoquarks (LQs) are hypothetical colour-triplet particles that carry both baryon and lepton quantum numbers ($B \neq 0, L \neq 0$). LQs couple simultaneously to both quarks and leptons at tree level, enabling direct transitions between the two. The spin of a LQ state is either 0 (scalar LQ) or 1 (vector LQ). Because of their SU(3) and SU(2) charge (colour and weak isospin, respectively), LQs can mediate flavour-changing neutral currents, and enable the violation of lepton flavour universality, which has been suggested as an explanation of recent measurements of B-meson decays [6]. New-physics models involving LQs might also resolve several interesting physical phenomena observed in nature. For instance, LQs can be used to explain the origins of the neutrino masses [7], as well as the origins of CP violation, thereby explaining the observed matter/antimatter asymmetry in the universe. In addition, LQs could provide a satisfying connection between the apparent symmetry of lepton and quark generations, as well as unification of the electromagnetic and weak forces at high energy.

The aim of the present analysis is to build an experimental pipeline that has a high sensitivity towards detecting the asymmetric pair production of scalar leptoquarks at high-energy proton-proton collisions in the $2\ell SS1\tau$ chanel, which stands for events with final state containing two light leptons of same sign and one hadronically decaying τ lepton. In this production process, two letpoquarks that are not charge conjugates of each other are produced simultaneously. Only scalar LQs are considered in the analysis, because the production phenomenon being analysed has not been studied for vector LQs. The analysis compares recorded data and Monte-Carlo (MC) samples in dedicated control regions and uses MC simulation data to train a machine learning model to separate signal and background. This classification is used to improve the signal and background separation in the Signal Region. Effective mass and signal probability are used to perform a profile likelihood fit to the asimov data in the signal region to extract Expected 95% CL Upper limits. The limits extracted from the two variables are compared and an expected exclusion on the leptoquark mass is established.

1.1 The $2\ell SS1\tau$ Channel

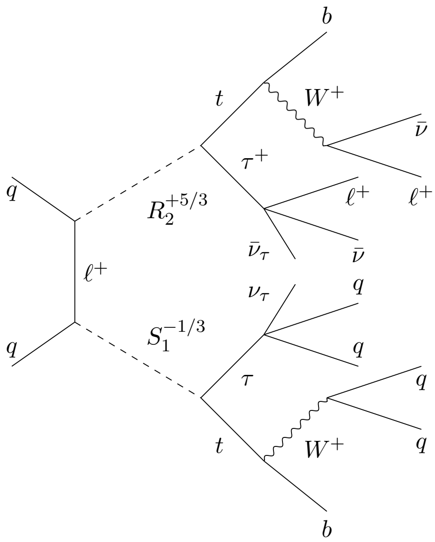


Figure 1.1: Asymmetric leptoquark pair production leading to $2\ell SS1\tau$ final state

The $2\ell SS1\tau$ channel stands for two light leptons of same sign and a hadronically decaying tau lepton in the final state. In the asymmetric pair proudction of leptoquarks this final state can be achieved if both the leptoquarks decay into a τ-lepton and a top-quark, and one of the top quark and τ lepton pair decays exclusively to jets while the other pair decays leptonically. One of the Feynman diagrams of the whole process leading to the $2\ell SS1\tau$ final state for the S_1 and R_2 scalar leptoquarks is given in Figure 1.1. This channel has been used to analyse rare processes like $t\bar{t}H(\tau\tau)$ and $t\bar{t}W$ [8] and it has been shown that $t\bar{t}H$ process shows more sensitivity in this channel compared to other multilepton channels [9]. Therefore, any results from Higgs boson searches can be reinterpreted as limits on leptoquarks in the $2lSS1\tau$ channel [10].

2 Theory

2.1 Quantum Field Theory

Quantum Field Theory is a field theory that incorporates the principles of quantum mechanics and special relativity. The Standard Model of Particle Physics is based on this theory where the particles are modeled as energy fluctuations in corresponding quantum fields. The basic idea of quantum field theory is that fields should be lorentz invariant and should be quantized based on the principles of quantum mechanics [11].

2.1.1 Lorentz Invariance

Lorentz invariance is a new kind of symmetry that emerges when we encounter phenomena close to speed of light. A system is considered lorentz invariant if it is symmetric under the Lorentz group. It can be considered as a generalization of rotation group which also contains boosts.

A transformation Λ is considered a lorentz transformation if it satisfies the following equation

$$\Lambda^T \eta \Lambda = \eta, \tag{2.1}$$

where η is the Minkowski metric and

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{2.2}$$

Considering the example of the simplest lorentz invariant equation:

$$\Box \phi(x) = 0 \tag{2.3}$$

The most general solution to this equation can be written as a fourier decomposition of plane waves with all possible modes.

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} (a_p e^{-ipx} + a_p^\star e^{ipx}) \tag{2.4}$$

2.1.2 Second Quantization

It is seen when the condition of lorentz invariance is imposed on the fields, plane waves automatically come into the discussion. A plane wave is also a solution to the equation of motion of a simple harmonic oscillator with angular frequency $\omega = |\vec{p}|$. In quantum mechanics a simple harmonic oscillator is quantized using the creation and annihilation operators: $a^\dagger, a$. We can similarly quantize each mode of the plane wave in equation 2.4 using the creation and annihilation operators for each mode: $a_p^\dagger, a_p$. Thus, the quantized hamiltonian for the minimal lorentz invariant equation can be written as

$$H_0 = \int \frac{d^3p}{(2\pi)^3}\omega_p \left(a_p^\dagger a_p + \frac{1}{2}\right) \tag{2.5}$$

We can also generalize the commutation relation of the creation and annihilation operators as follows.

$$[a, a^\dagger] = 1 \longrightarrow [a_k, a_p^\dagger] = (2\pi)^3\delta^3(\vec{p} - \vec{k}) \tag{2.6}$$

The $a_p^\dagger$ operators create particles with momentum p

$$a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |\vec{p}\rangle \tag{2.7}$$

Where $|0\rangle$ is the vacuum state and $|\vec{p}\rangle$ is a single particle state with momentum $\vec{p}$. The vacuum state satisfies the following normalization condition.

$$\langle 0|0\rangle = 1 \tag{2.8}$$

$$\implies \langle \vec{p}|\vec{k}\rangle = 2\omega_p(2\pi)^3\delta^3(\vec{p} - \vec{k}) \tag{2.9}$$

And the indentity of the one-particle momentum state can be given as

$$\mathbb{1} = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\omega_p} |\vec{p}\rangle \langle \vec{p}| \tag{2.10}$$

In terms of the annihilation and creation operators, a free quantum scalar field can be written as an operator.

$$\phi_0(\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{(2\omega_p)}(a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}}) \tag{2.11}$$

2.2 Dirac Equation

The dynamics of all matter particles (fermions) is governed by the dirac equation, which is given as follows [12].

$$(i\partial\!\!\!/ - m)\psi = 0 \qquad , \partial\!\!\!/ \equiv \gamma^\mu\partial_\mu \tag{2.12}$$

where γ^μ are dirac matrices. This is a lorentz invariant equation and can be written as a lorentz invariant lagrangian.

$$\mathcal{L} = \overline{\psi}(i\partial\!\!\!/ - m)\psi \tag{2.13}$$

ψ represents the free particle fermion state and is a 4×1 column matrix and is called a Dirac Spinor. The dirac gamma matrices satisfy the following equations

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \tag{2.14}$$

$$\gamma^{0\dagger} = \gamma^0 \;\&\; \gamma^{k\dagger} = -\gamma^k \tag{2.15}$$

The physical content of the dirac equation is not governed by the explicit representation of the gamma matrices but by equations 2.14 and 2.15.

2.2.1 Interactions

Interaction in quantum field theory are introduced as consequences of imposing local gauge symmetries in the system. The simplest one is the $U(1)$ symmetry of electromagnetism. By imposing the U(1) local gauge symmetry in the dirac lagrangian we are forced to introduce a new vector field A^μ, with a definite transformation property under the $U(1)$ local gauge transformation, this field has the same transformation property as that of the vector potential from electromagnetism. And therefore, after imposing the local $U(1)$ symmetry and adding another term for the dynamics of A^μ we get the QED lagrangian, which governs the dynamics of matter particles and QED vector boson, the photon.

$$\mathcal{L} = \overline{\psi}(i\partial\!\!\!/ - m)\psi + e\overline{\psi}A\!\!\!/\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{2.16}$$

where $F_{\mu\nu}$ is the electromagnetic field tensor.

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{2.17}$$

This is a maifestly lorentz invariant lagrangian which also respects the local $U(1)$ symmetry.

The weak and the strong interactions are also introduced by imposing the $SU(2)$ and $SU(3)$ local gauge symmetries respectively and introducing the lorentz invariant dynamic terms for all the new fields. These new fields correspond to spin 1 vector boson fields which are responsible for carrying the corresponding forces. According to Noether's Theorem, each local gauge symmetry introduced in the system corresponds to a conserved charge of the system, the conserved charges for each of the interactions is discussed in section 2.3.

2.3 The Standard Model of Particle Physics

The Standard Model of Particle Physics is a renormalizable quantum field theory of gauge symmetries $SU(3)_C \times SU(2)_L \times U(1)_Y$ which is spontaneously broken by the vacuum expectation value of the Higgs field to $SU(3)_C \times U(1)_{EM}$. Each symmetry group corresponds to a fundamental interaction and the generators of the symmetry groups correspond to the vector bosons responsible for carrying that interaction. $SU(3)_C$ corresponds to strong interaction which has 8 gluons corresponding to each generator, $SU(2)_L$ and $U(1)_Y$ correspond to electroweak interactions with their 3+1 generators for 4 vector bosons of the electroweak sector. After electroweak symmetry breaking, three of the four vector bosons of electrweak sector acquire mass ($W^{\pm}, Z$), and one remains massless (the photon,γ). The conserved charge of each of these interactions is the colour charge of $SU(3)_C$, weak isospin of $SU(2)_L$ and hypercharge corrsponding to $U(1)_Y$. The subscript L in $SU(2)_L$ signifies that only left handed particles are affected by this symmetry.

The full particle content of the standard model can be divided mainly into two parts based on the spin of the particles: fermions, which are spin 1/2 particles, they make up the matter particles and bosons, which are integer spin particles. Fermions can further be divided into quarks, which carry colour charge and leptons, that do not carry colour charge. Fermions are observed to come in three generations with each generation containing 2 quarks and 2 leptons and subsequent generations consisting of heavier particles. Bosons can further be divided into 2 catogories: vector bosons, spin 1 force carrying particles discussed in the previous paragraph and the spin 0 scalar particle, the Higgs boson. The full particle content of the Standard Model is given in Figure 2.1.

2.3.1 Quantum Electrodynamics

The theory that describes the electromagnetic interactions is quantum electrodynamics. As described in section 2.2.1, it is a consequence of introducing $U(1)$ local gauge symmetry and the QED lagrangian is given by

$$\mathcal{L} = \overline{\psi}(i\partial\!\!\!/ - m)\psi + e\overline{\psi}A\!\!\!/\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{2.18}$$

where the first and the last terms are kinetic terms of fermion and the photon, the second term is the mass term of fermion and the third term is the interaction term that governs how the fermions and photons interact.

2.3.2 Qunatum Chromodynamics

Quantum Chromodynamics is the theory of strong interactions and is introduced as a consequence of $SU(3)_C$ local gauge symmetry. Unlike QED, this is a non-abelian gauge

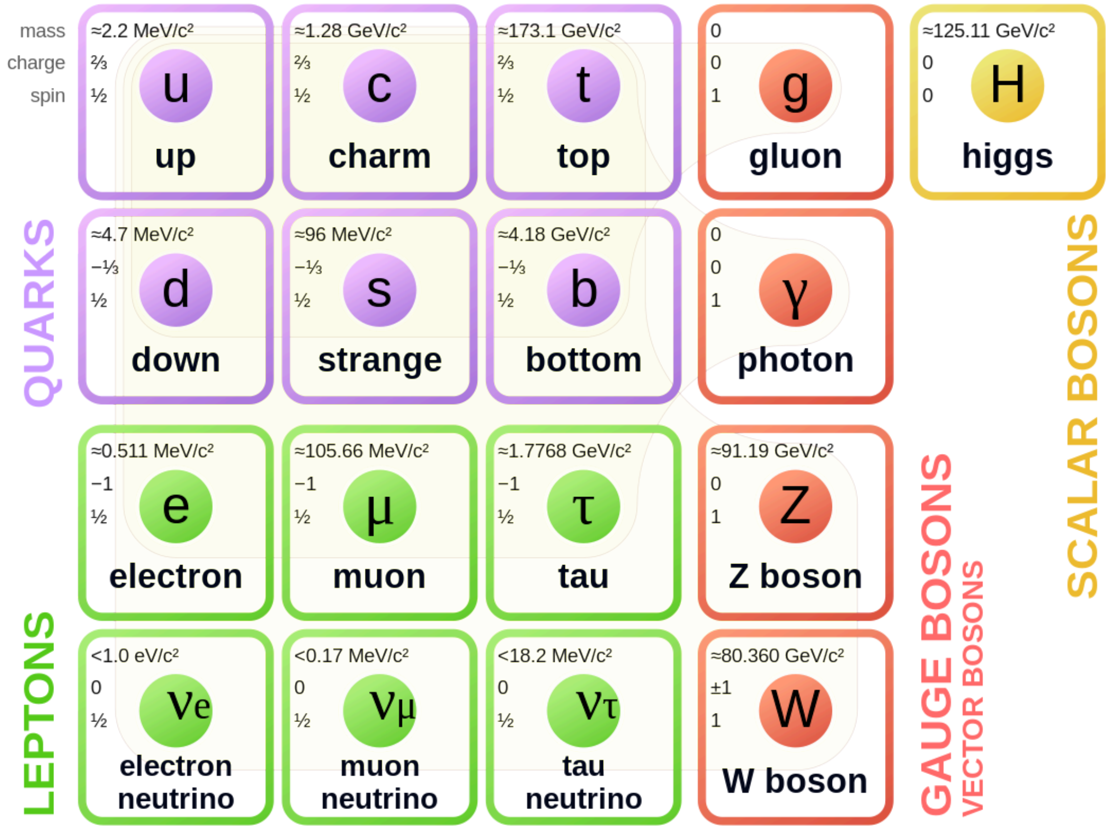


Figure 2.1: Particle content of the standard model [13]

theory. The QCD lagrangian can be given by

$$\mathcal{L} = \overline{\psi}(i\partial\!\!\!/ - m)\psi + g_s\overline{\psi}\gamma^\mu G^a_\mu T_a\psi - \frac{1}{4}G^a_{\mu\nu}G^{\mu\nu}_a, \tag{2.19}$$

where G^a_μ are vector fields corresponding to gluons, T_a are the eight generators of $SU(3)$ and $G^a_{\mu\nu}$ is the gluon field strength tensor

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f_{abc} G^b_\mu G^c_\nu, \tag{2.20}$$

where f_{abc} is the structure constant of the group $SU(3)$ and defines the commutation relations of the generators

$$[T_a, T_b] = i f^c_{ab} T_c. \tag{2.21}$$

2.3.3 The Electroweak Standard Model

Similar to QED and QCD the weak interactions are introduced as a consequence of $SU(2)_L$ local gauge symmetry of the lagrangian. Like QCD, the theory of weak interactions is also a non-abelian gauge theory which only couples to the left handed particle states. The lagrangian after introducing $SU(2)_L$ local gauge symmetry can be written as:

$$\mathcal{L} = i\overline{\psi}\partial\!\!\!/\psi + g\overline{\psi}T_k\gamma^\mu W_\mu^k\psi - \frac{1}{4}W_{\mu\nu}^k W_k^{\mu\nu} \tag{2.22}$$

Note that after introducing $SU(2)_L$ symmetry, the fermion mass term cannot be directly added to the lagrangian as it violates the $SU(2)_L$ symmetry. The physical $W^\pm$ bosons are defined as the linear combination of W^1 and W^2.

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \tag{2.23}$$

In standard model, the electromagnetic interactions are not directly introduced as a consequence of $U(1)_{EM}$ symmetry. Instead, $U(1)_Y$ symmetry is introduced which corresponds to the conserved quantity called hypercharge. The gauge field B_μ introduced by this symmetry and the W^3 field of $SU(2)_L$ are used to construct the physical Z-boson and photon fields

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} \tag{2.24}$$

where θ_W is the Weinberg angle.
This is called Electroweak Unification and a consequence of this is that the physical couplings of the electroweak sector mix with each other.

2.3.4 Spontaneous Symmetry Breaking

Spontaneous symmetry breaking is an essential component of the Standard Model of Particle Physics. The fundamental interactions are introduced as a consequence of local gauge symmetries of the lagrangian, because of this it is not possible to directly introduce mass terms for any particle as this would violate the local gauge invariance. Consider the example of an electron and a photon. The electron mass term can be written as

$$\mathcal{L}_{e_m} = -m_e\bar{e}e = -m_e(\bar{e}_R e_L + \bar{e}_L e_R) \tag{2.25}$$

Since the $SU(2)_L$ transformation applies only to the left handed particle states, this term is not invaraint under $SU(2)_L$ transformations. Similarly, since the photon field tranforms as $A_\mu \to A'_\mu = A_\mu - \partial_\mu \chi$ under local $U(1)$ transformation, its corresponding mass term is not invariant

$$\mathcal{L}_{A_m} = \frac{1}{2} m_\gamma^2 A_\mu A^\mu \to \frac{1}{2} m_\gamma^2 (A_\mu - \partial_\mu \chi)(A^\mu - \partial^\mu \chi) \tag{2.26}$$

Spontaneous symmetry breaking is a mechanism in which extra massless scalar fields are introduced in the lagrangian with certain form of symmetric potential. This potential is defined in such a way that the vacuum states are degenerate and the choice of the vacuum state breaks the symmetry of the lagrangian. As a result of choosing this vacuum state, the lagrangian can be written in terms of a massive scalar field and goldstone boson fields, which contribute to providing masses to the vector bosons of gauge symmetries.

In the Standard Model, the scalar field introduced is called the Higgs field and it is introduced as an $SU(2)_L$ doublet to make sure that it only provides mass to the bosons of the weak sector and the photon remains massless

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{2.27}$$

The lagrangian for this complex scalar field doublet is defined as

$$\mathcal{L}_{Higgs} = (\partial \Phi)^\dagger (\partial \Phi) - V(\Phi) \qquad , V(\Phi) = \mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \tag{2.28}$$

for $\mu^2 < 0$, the potential has an infinite set of minima

$$\Phi^\dagger \Phi = \frac{v^2}{2} = -\frac{\mu^2}{2\lambda} \tag{2.29}$$

Since the neutral photon is required to be massless after symmetry breaking, the vacuum expectation value of the Higgs field should be defined as

$$\langle 0 | \Phi | 0 \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} . \tag{2.30}$$

After spontaneous symmetry breaking, there remains a massive scalar field and three goldstone bosons which give the longitudinal degrees of freedom to the $W^\pm$ and Z bosons. In the unitary gauge the Higgs doublet is wirtten as

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} , \tag{2.31}$$

where $h(x)$ corresponds to the field of the physical Higgs boson.

2.3.5 The Standard Model Lagrangian

The full standard model lagrangian is given by

$$\begin{aligned}\mathcal{L} = &\underbrace{\bar{Q}_i \not{\mathcal{D}}_\mu Q^i + \bar{L}_i \not{\mathcal{D}}_\mu L^i + \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{4}G^a_{\mu\nu}G^{\mu\nu}_a + \frac{1}{4}W^k_{\mu\nu}W^{\mu\nu}_k}_{\mathcal{L}_{kin}+\mathcal{L}_{int}} \\ &+ \underbrace{(\mathcal{D}_\mu\Phi)^\dagger(\mathcal{D}_\mu\Phi) - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2}_{\mathcal{L}_{Higgs}} \\ &- \underbrace{y^q_{ij}\bar{Q}^i_L\Phi q^j_R - y^q_{ij}\bar{Q}^i_L\Phi^C q^j_R - y^\ell_{ij}\bar{L}_L\Phi\ell_R + h.c.}_{\mathcal{L}_{Yukawa}},\end{aligned} \tag{2.32}$$

$$\mathcal{D}_\mu \equiv \partial_\mu - ig'\frac{Y}{2}B_\mu - igW^k_\mu T_k - ig_s G^a_\mu \mathcal{T}_a. \tag{2.33}$$

where $i, j = 1, 2, 3$ run over all generations and the subscript L and R correspond to the chirality of the particles.

2.4 Leptoquarks

Leptoquarks are hypothetical particles that couple a lepton and a quark at tree level. These particles are predicted by many BSM theories and are associated with Grand Unified Theories with SU(5) or SO(10) Symmetry Groups that unify quarks and leptons [14–18]. They can be of scalar (spin 0) or vector (spin 1) nature and are assigned a Fermion number defined as follows

$$F = 3B + L, \tag{2.34}$$

where B and L are Baryon and Lepton numbers. A quark (lepton) is defined to carry a baryon (lepton) number of 1/3 (1). In total, there are six scalar leptoquark states ($S_3, R_2, \tilde{R}_2, \tilde{S}_1, S_1, \bar{S}_1$) and six vector leptoquark states ($U_3, V_2, \tilde{V}_2, \tilde{U}_1, U_1, \bar{U}_1$) [19].

The present analysis focuses on what is called the asymmetric pair production of scalar leptoquarks. In this channel the leptoquarks being produced are not charge conjugates of each other, therefore, it is necessary that the leptoquarks being produced couple to the same pair of quark and lepton. Such a scenario is possible in the case of S_1 and R_2 leptoquarks and hence this analysis will focus on only these two leptoquarks.

2.4.1 R_2 Lagrangian

The renormalizable yukawa coupling terms for the R_2 leptoquark are given as follows

$$\mathcal{L} \supset -y_{2\,ij}^{RL}\bar{u}^i_R R^a_2 \epsilon^{ab} L^{j,b}_L + y_{2\,ij}^{LR}\bar{e}^i_R R^{a\,*}_2 Q^{j,a}_L + h.c., \tag{2.35}$$

$(SU(3)_C, SU(2)_L, U(1)_Y)$	Spin	Symbol	Type	F
$(\mathbf{3}, \mathbf{3}, 1/3)$	0	S_3	LL	-2
$(\mathbf{3}, \mathbf{2}, 7/6)$	0	R_2	RL,LR	0
$(\mathbf{3}, \mathbf{2}, 1/6)$	0	$\tilde{R}_2$	RL	0
$(\mathbf{3}, \mathbf{1}, 4/3)$	0	$\tilde{S}_1$	RR	-2
$(\mathbf{3}, \mathbf{1}, 1/3)$	0	S_1	LL,RR	-2
$(\mathbf{3}, \mathbf{3}, 2/3)$	1	U_3	LL	0
$(\mathbf{3}, \mathbf{2}, 5/6)$	1	V_2	RL,LR	-2
$(\mathbf{3}, \mathbf{2}, -1/6)$	1	$\tilde{V}_2$	RL	-2
$(\mathbf{3}, \mathbf{1}, 5/3)$	1	$\tilde{U}_1$	RR	0
$(\mathbf{3}, \mathbf{1}, 2/3)$	1	U_1	LL,RR	0

Table 2.1: List of scalar and vector leptoquarks [19]

where y_2^{RL} and y_2^{LR} are 3×3 yukawa coupling matrices, u_R^i and e_R^i are right handed up-type quark and charged lepton singlets of ith generation and L_L^j and Q_L^j are left handed lepton and quark $SU(2)$ doublets of jth generation. a, b are $SU(2)$ indices.

2.4.2 S_1 Lagrangian

The renormalizable yukawa coupling terms for the S_1 leptoquark are given as follows

$$\mathcal{L} \supset +y_{1\,ij}^{LL} \bar{Q}_L^{C\,i,a} S_1 \epsilon^{ab} L_L^{j,b} + y_{1\,ij}^{RR} \bar{u}_R^{C\,i} S_1 e_R^j + \cdots + h.c., \tag{2.36}$$

where y^{LL} and y^{RR} are yukawa coupling matrices and "^{C}" denotes charge conjugation. There are other terms in the S_1 lagrangian but these are either quark-quark couplings or couplings to right handed neutrino fields which are irrelevant for the present analysis and hence have been left out.

2.4.3 Conventional Leptoquark Production

Searches targeting leptoquarks at the LHC have till now focused on single production or conventional pair production, in which the leptoquarks being produced are charge conjugates of each other. The Feynman diagrams corresponding to both production mechanisms are given in Figures 2.2 and 2.3.

Since, single production contains one lepton-quark-leptoquark vertex the production cross section for this process can be given by

$$\sigma_{single} = f(m_{LQ}) |y_i|^2, \tag{2.37}$$

Figure 2.2: Single production

where f is some function of leptoquark mass and y_i is the Yukawa coupling of lepton-quark-leptoquark vertex.

Conventional pair production can proceed in two ways, either by the QCD driven process given in the left side of Figure 2.3 or by the t-channel lepton exchange process given in the right side of Figure 2.3. The cross section for conventional pair production is given in eq. 2.38.

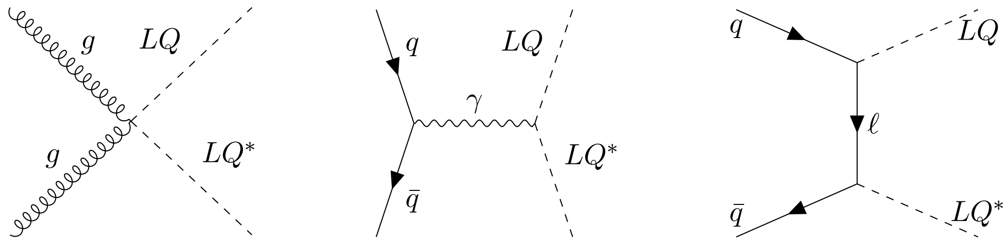


Figure 2.3: Conventional pair production

$$\sigma_{pair} = f_{QCD}(m_{LQ}) + f_{int}(m_{LQ})|y_i|^2 + f_{t-chan}(m_{LQ})|y_i|^4 \tag{2.38}$$

The cross section comprises of a QCD driven term and a t-channel lepton exchange term which depends on $|y_i|^4$ because of the presence of two lepton-quark-leptoquark vertices. The center term in the equation is the interference term between the QCD driven term and the t-channel lepton exchange.

2.4.4 Asymmetric Pair Production

The asymmetric pair production of leptoquarks is a new physics scenario that is initiated by a quark-quark initial state as opposed to a quark-antiquark initial state in the conventional pair production [20]. This new production channel has the potential to be more important than the dominant QCD driven pair production and single leptoquark production because of PDF suppression and a more pronounced scaling with respect to Yukawa couplings. Figure 2.4 shows an illustration of asymmetric pair production process.

Due to conservation of fermion number, the final state of this production process must have one leptoquark with $F = 0$ and the other with $F = |2|$. Therefore, in the final state, one of the leptoquarks should be $S_3, \tilde{S}_1, S_1$ or $\bar{S}_1$ and the other shuld be either R_2 or $\tilde{R}_2$. The cross-section of this process can be given by

$$\sigma_{q_1 q_2} = a_{q_1 q_2}(m_{LQ_1}, m_{LQ_2})|y_{q_1} y_{q2}|^2, \tag{2.39}$$

where the first Yukawa coupling is the coupling of LQ_1 to q_1 and the second one is of LQ_2 to q_2.

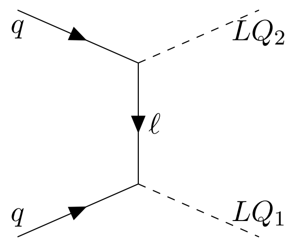


Figure 2.4: Asymmetric pair production

This analysis only focuses on the asymmetric pair production of scalar leptoquarks $S_1^{+1/3}$ and $R_2^{+5/3}$, therefore the relevant lagrangian terms are

$$\mathcal{L} \supset +(y_1 V^\dagger)_{ij} \bar{e}_R^i u_L^j R^{+5/3\,*} + y_{2\,ij} \bar{u}_R^{C\,i} S_1^{+1/3} e_R^j + h.c., \tag{2.40}$$

where V is the CKM matrix.

3 The Large Hadron Collider and the ATLAS Detector

3.1 The Large Hadron Collider

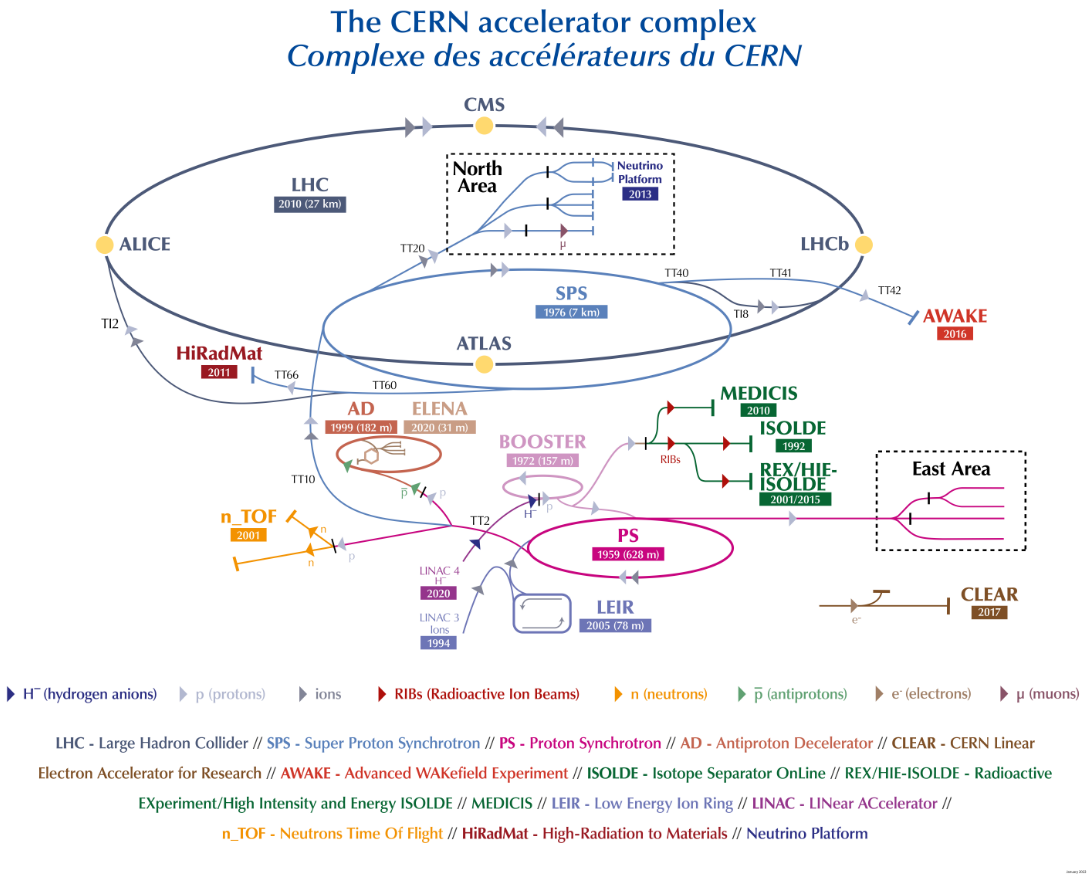


Figure 3.1: CERN accelerator complex [21]

The Large Hadron Collider (LHC) [22] is the world's largest and most powerful particle accelerator that pushes protons or ions to near the speed of light. It first started on 10 September 2008, and remains the latest addition to CERN' s accelerator complex.

The LHC consists of a 27-kilometer ring of superconducting magnets to boost the energy of the particles along the way. The accelerator sits in a tunnel 100 metres underground at CERN, the European Organization for Nuclear Research, on the Franco-Swiss border near Geneva, Switzerland.

Inside the accelerator, two high-energy particle beams travel at close to the speed of light before they are made to collide. The beams travel in opposite directions in separate beam pipes. They are guided around the accelerator ring by a strong magnetic field maintained by superconducting electromagnets. The electromagnets are built from coils of special electric cable that operates in a superconducting state, efficiently conducting electricity without resistance or loss of energy. This requires chilling the magnets to -271.3°C. For this reason, much of the accelerator is connected to a distribution system of liquid helium, which cools the magnets, as well as to other supply services.

Thousands of magnets of different varieties and sizes are used to direct the beams around the accelerator. These include 1232 dipole magnets, 15 metres in length, which bend the beams, and 392 quadrupole magnets, each 5–7 metres long, which focus the beams. Just prior to collision, another type of magnet is used to “squeeze” the particles closer together to increase the chances of collisions.

3.2 The ATLAS Detector

The ATLAS detector [23] is one of the major particle detectors constructed at the Large Hadron Collider (LHC) at CERN. It sits at one of the eight interaction points in the LHC and was Designed to explore a wide range of particle physics phenomena. ATLAS plays a critical role in advancing our understanding of the fundamental constituents and forces of nature. The detector is situated in a cavern 100 meters below the ground near Geneva, Switzerland. It has a cylindrical structure and spans approximately 46 meters in length, 25 meters in diameter and weighs about 7000 tonnes, making it one of the largest volume detectors ever constructed for a particle collider.

The primary objective of the ATLAS detector is to perform precision measurements and test the Standard Model. It also addresses pivotal questions in modern physics, including the origins of mass and the nature of dark matter. Notably, the detector was instrumental in the discovery of the Higgs boson in 2012, a breakthrough that confirmed the existence of the last predicted component of the Standard Model of particle physics.

The ATLAS detector is a many-layered instrument designed to detect some of the tiniest yet most energetic particles ever created on earth. It consists of four main detecting subsystems wrapped concentrically in layers around the collision point to record the trajectory, momentum, and energy of particles, allowing them to be individually identified and measured. A huge magnet system bends the paths of the charged particles so that their momenta can be measured as precisely as possible. Over a billion particle interactions take place in the ATLAS detector every second and only one in a million collisions is flagged as potentially interesting and recorded for further study.

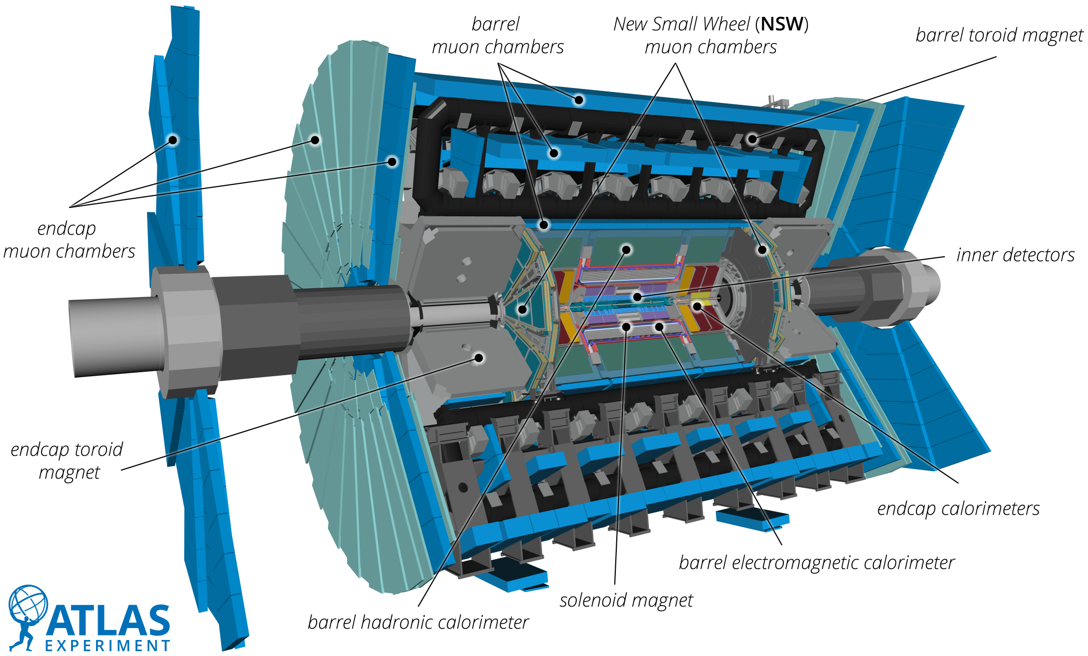


Figure 3.2: The ATLAS detector [24]

The 4 main sub-detectors of ATLAS are:

- Inner Detector
- Calorimeter
- Magnetic System
- Muon Spectrometer

3.2.1 Inner Detector

It is the first part of ATLAS to see the decay products of the collisions. It is very compact and highly sensitive. It consists of three different systems of sensors all immersed in a magnetic field parallel to the beam axis. The Inner Detector measures the direction, momentum, and charge of electrically-charged particles produced in each proton-proton collision. The main components of the Inner Detector are: Pixel Detector, Semiconductor Tracker (SCT), and Transition Radiation Tracker (TRT).

- **Pixel Detector:** Located just 3.3 cm from the LHC beam line, the Pixel Detector is the first point of detection in the ATLAS experiment. It is made up of four layers of silicon pixels, with each pixel smaller than a grain of sand. As charged particles burst out from the collision point, they leave behind small energy deposits in the Pixel Detector. These signals are measured with a precision of almost 10 μm to determine the origin and momentum of the particle. The Pixel Detector is incredibly compact, with over 92 million pixels and almost 2000 detector elements.

- **Semiconductor Tracker:** The Semiconductor Tracker surrounds the Pixel Detector and is used to detect and reconstruct the tracks of charged particles produced during collisions. It consists of over 4,000 modules of 6 million micro-strips of silicon sensors. Its layout is optimised such that each particle crosses at least four layers of silicon. This allows scientists to measure particle tracks with a precision of up to 25 μm.

- **Transition Radiation Tracker:** The third and final layer of the Inner Detector is the Transition Radiation Tracker (TRT). Unlike its neighbouring sub-detectors, the TRT is made up of 300,000 thin-walled drift tubes. Each tube is just 4 mm in diameter, with a 30 μm gold-plated tungsten wire in its centre. The straws are filled with a gas mixture. As charged particles cross through the straws, they ionise the gas to create a detectable electric signal. This is used to reconstruct their tracks and, owing to the so-called transition radiation, provides information on the particle type that flew through the detector, i.e. if it is an electron or pion.

3.2.2 Calorimeter

The Calorimeter measures the energy a particle loses as it passes through the detector. They are designed to absorb most of the particles coming from a collision, forcing them to deposit all of their energy and stop within the detector. ATLAS calorimeters consist of layers of an “absorbing” high-density material that stops incoming particles, interleaved with layers of an “active” medium that measures their energy.

Electromagnetic calorimeters measure the energy of electrons and photons as they interact with matter. Hadronic calorimeters sample the energy of hadrons (particles that contain quarks, such as protons and neutrons) as they interact with atomic nuclei.

Calorimeters can stop most known particles except muons and neutrinos.

The components of the ATLAS calorimetry system are:

- **The Liquid Argon (LAr) Calorimeter:** The Liquid Argon (LAr) Calorimeter surrounds the ATLAS Inner Detector and measures the energy of electrons, photons and hadrons. It features layers of metal (either tungsten, copper or lead) that absorb incoming particles, converting them into a shower of new, lower energy particles. These particles ionise liquid argon sandwiched between the layers, producing an electric current that is measured. By combining all of the detected currents, physicists can determine the energy of the original particle that hit the detector. The central region of the calorimeter is specially designed to identify electrons and photons. It features a characteristic accordion structure, with a honeycomb pattern, to ensure that no particle escapes unchallenged. To keep the argon in liquid form, the calorimeter is kept at $-184°$ C. Specially-designed, vacuum-sealed cylinders of cables bring the electronic signals from the cold liquid argon to the warm area where the readout electronics are located.
- **The Tile Hadronic Calorimeter:** The Tile Calorimeter surrounds the LAr calorimeter and measures the energy of hadronic particles, which do not deposit all of their energy in the LAr Calorimeter. It is made of layers of steel and plastic scintillating tiles. As particles hit the layers of steel, they generate a shower of new particles. The plastic scintillators in turn produce photons, which are converted into an electric current whose intensity is proportional to the original particle’s energy. The Tile Calorimeter is made up of about 420,000 plastic scintillator tiles working in sync. It is the heaviest part of the ATLAS experiment, weighing almost 2900 tonnes.

3.2.3 Magnetic System

The Magnetic System bends particles around the various layers of detector systems. By bending the trajectories of charged particles, ATLAS can measure their momentum and charge. This is done using two different types of superconducting magnet systems – solenoidal and toroidal. These impressive systems are cooled to about 4.5 K (–268° C) in order to provide the necessary strong magnetic fields.

The main sections of the magnet system are:

- **Central Solenoid Magnet:** The ATLAS solenoid surrounds the inner detector at the core of the experiment. This powerful magnet is 5.8 m long, 2.56 m in diameter and weighs over 5 tonnes. It provides a 2 Tesla magnetic field in just 4.5 cm thickness. This is achieved by embedding over 9 km of niobium-titanium superconductor wires into strengthened, pure aluminum strips, thus minimising possible interactions between the magnet and the particles being studied.

- **Barrel Toroid and End-cap Toroids:** The ATLAS toroids use a series of eight coils to provide a magnetic field of up to 3.5 Tesla, used to measure the momentum of muons. There are three toroid magnets in ATLAS: two at the ends of the experiment, and one massive toroid surrounding the centre of the experiment. At 25.3 m in length, the central toroid is the largest toroidal magnet ever constructed. It uses over 56 km of superconducting wire and weighs about 830 tonnes. The end-cap toroids extend the magnetic field to particles leaving the detector close to the beam pipe. Each end-cap is 10.7 m in diameter and weighs 240 tonnes.

3.2.4 Muon Spectrometer

The outer layer of the ATLAS experiment is made of muon detectors. They identify and measure the momenta of muons. Five different detector technologies are used: Thin Gap Chambers, Resistive Plate Chambers, Monitored Drift Tubes, Small-Strip Thin-Gap Chambers and Micromegas.

- **Precision Detectors:** The precision detectors of the Muon Spectrometer can measure the position of a muon with an accuracy better than one tenth of a millimetre. They are made of Monitored Drift Tube (MDT) detectors, which consist of aluminium tubes, 3 cm in diameter, filled with a suitable gas mixture. When a muon passes through a tube, it ionises the gas, releasing electrons. These electrons drift toward a thin wire stretched along the centre of the tube, where they generate an electrical signal. By stacking more than 380,000 such tubes in multiple layers, the spectrometer can accurately reconstruct the trajectory of muons.

- **Fast-Response Detectors:** ATLAS relies on fast-response detectors to rapidly identify collision events that are potentially interesting for further physics analysis. This decision is made within $2.5\,\mu$s. In the central region of the detector, Resistive Plate Chambers (RPCs) are used, which consist of pairs of parallel plastic plates separated by a gas gap and held at an electric potential difference. At the ends of ATLAS, Thin Gap Chambers (TGCs) are installed, composed of arrays of parallel $30\,\mu$m wires in a gas mixture. Muons ionise the gas as they traverse these chambers, producing signals that can be detected. To cope with the very high particle flux close to the beamline, ATLAS also employs Micromegas and Small-Strip Thin Gap Chambers (sTGCs). These advanced detectors provide fast and precise muon tracking even in high-density environments. By combining information from all of these fast-response detectors, ATLAS achieves a coarse measurement of the muon momentum. This enables the trigger system to decide whether a collision event should be kept for detailed analysis or discarded.

3.3 Trigger and Data Acquisition System

The Trigger and Data Acquisition System selects events with distinguishing characteristics that make them interesting for physics analyses. ATLAS is designed to observe up to 1.7 billion proton-proton collisions per second, with a combined data volume of more than 60 million megabytes per second. However, only some of these events will contain interesting characteristics that might lead to new discoveries. The Trigger and Data Acquisition system ensures optimal data-taking conditions and selects the most interesting collision events for study.

The ATLAS trigger system carries out the selection process in two stages.

- The first-level hardware trigger, constructed with custom-made electronics located on the detector, works on a subset of information from the calorimeters and the Muon Spectrometer. The decision to keep the data from an event is made less than 2.5 microseconds after the event occurs. During this time the event data is kept in storage buffers. If the event is selected it is passed on to the second-level trigger, which can accept up to 100,000 events per second.
- The second-level software trigger operates from a large farm of about 40,000 CPU cores. In just 200 microseconds, it conducts very detailed analyses of each collision event, examining data from specific detector regions. The second-level trigger finally selects about 1000 events per second and passes them on to a data storage system for offline analysis.

4 Statistical Data Analysis in High-Energy Physics

The extraction of meaningful physics results from collider data requires a careful combination of statistical data analysis, modern machine learning methods, and dedicated computational frameworks. In high-energy physics (HEP), statistical inference provides the foundation for discovery and exclusion: new phenomena are established through hypothesis testing, while in the absence of significant evidence, upper limits on potential signals are set. At the same time, recent advances in machine learning have opened new opportunities for more efficient and accurate event classification, surpassing the performance of traditional cut-based analyses. To implement these techniques in practice, specialized frameworks have been developed by experimental collaborations to streamline event generation, object reconstruction, and statistical fitting.

This chapter introduces the statistical methodology underlying hypothesis testing in particle physics, the role of machine learning techniques in improving signal–background discrimination, and the computational frameworks used in this thesis. Together, these tools form the backbone of the analysis chain used to probe physics beyond the Standard Model (BSM).

4.1 Hypothesis Testing

Hypothesis Testing is used to determine whether the observed data is compatible with a given hypothesis. Two competing hypotheses are defined: A null hypothesis (H_0) and an alternative hypothesis (H_1). The result of a hypothesis test is not the establishment of the alternate hypothesis, the result is just an evidence against the null hypothesis. To quantify the compatibility of data with the null hypothesis, a test statistic is defined, which depends on the observed data. This test statistic is used to evaluate the p-value of the test, which is the probability of observing a data more incompatible with the null hypothesis than what is observed. If this p-value is observed to be less than some threshold, the null hypothesis is rejected.

In the context of particle physics data analysis, the hypothesis test is used in the search for BSM physics. Two competing hypotheses are defined: The background-only hypothesis, which includes all the Standard Model processes, and the signal+background hypothesis, where signal is the BSM process that the analysis focuses on. A general hypothesis test in particle physics involves two steps:

- Discovering a new signal process by gathering evidence against the background only hypothesis, if enough evidence is not obtained against the background only hypothesis then
- The analysis proceeds to set limits on the cross-section of the new signal process.

For the purpose of discovery, the background only hypothesis is regarded as the null hypothesis (H_0) and the signal+background hypothesis is regarded as the alternative hypothesis (H_1), while for limit setting the vice versa is considered [25].

Different test statistics can be used to test the compatibility of data with a hypothesis, in particle physics the negative log of profile likelihood ratio is often used. Assuming the parameter of interest (POI) is the signal strength (μ), $\boldsymbol{\theta}$ are the nuisance parameters and $L(\mu, \boldsymbol{\theta})$ is the likelihood function, then

$$t_\mu = -2ln\frac{L(\mu, \hat{\hat{\boldsymbol{\theta}}})}{L(\hat{\mu}, \hat{\boldsymbol{\theta}})}, \tag{4.1}$$

where μ is the hypothesized value of POI, $\hat{\hat{\boldsymbol{\theta}}}$ is the conditional maximum likelihood estimator of $\boldsymbol{\theta}$, and $\hat{\mu}$, $\hat{\boldsymbol{\theta}}$ are the unconditional maximum likelihood estimators of μ and $\boldsymbol{\theta}$. t_μ values close to zero correspond to good agreement between data and hypothesis.

The likelihood function is the Poisson probabilities for all the bins in the signal and the control region

$$L(\mu, \boldsymbol{\theta}) = \prod_{j=1}^{N} \frac{(\mu s_j + b_j)^{n_j}}{n_j!} e^{-(\mu s_j + b_j)} \prod_{k=1}^{M} \frac{u_k^{m_k}}{m_k!} e^{-u_k}, \tag{4.2}$$

where s_j and b_j are the expected number of signal and background events in the signal region in the j^{th} bin while n_j is the total number of observed events in the signal region in the jth bin and u_k and m_k are the expected and observed number of events in the control region in the k^{th} bin respectively, assuming a signal region and only one control region.

The p-value for a given hypothesis can be calculated using the test statistic

$$p_\mu = \int_{t_{\mu,obs}}^{\infty} f(t_\mu|\mu) dt_\mu, \tag{4.3}$$

where $t_{\mu,obs}$ is the value of t_μ observed from data and $f(t_\mu|\mu)$ is the probability distribution function (pdf) of t_μ assuming a signal strength of μ. For rejecting the background-only hypothesis a 5σ significance is considered appropriate which corresponds to $p = 2.87 \times 10^{-7}$, while for limit setting, $p = 0.05$ (95% Confidence Level) is used.

4.1.1 The CL_S Method

In case the background-only hypothesis cannot be rejected, upper limits are set on the signal strength. There are multiple ways of setting upper limits, one of the methods is called the CL_s procedure [26].

Assuming the test statitic used for comparing different hypotheses is q and its corresponding distribution under the signal+background and background only hypotheses are $f(q|s+b)$ and $f(q|b)$ respectively. Then assuming the b-only ditribution is shifted towards right(as in Fig 4.1), the p-value of the s+b hypthesis and b-only hypothesis can be given by

$$p_{s+b} = P(q \geq q_{obs}|s+b) = \int_{q_{obs}}^{\infty} f(q|s+b)dq \tag{4.4}$$

$$p_b = P(q \leq q_{obs}|b) = \int_{-\infty}^{q_{obs}} f(q|b)dq. \tag{4.5}$$

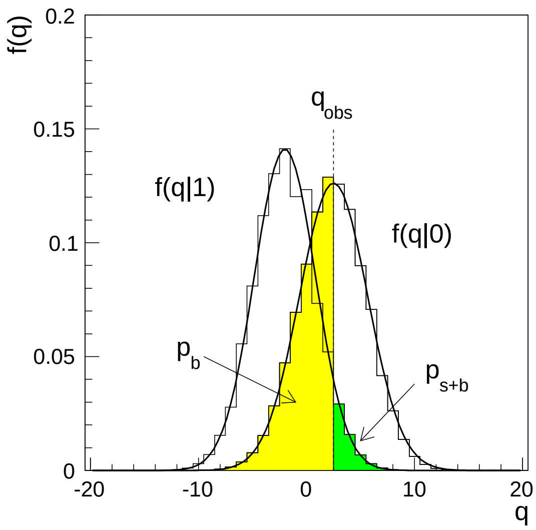


Figure 4.1: Distribution of q under s+b and b-only hypotheses [26]

In the CL_{s+b} approach, one performs a conventional statistical test of the signal+background (s+b) hypothesis by evaluating its corresponding p-value, p_{s+b}. The signal model is considered excluded at a confidence level of $1-\alpha = 95\%$ if the test yields a p-value satisfying the specified exclusion criterion.

$$p_{s+b} < \alpha \tag{4.6}$$

A confidence interval with confidence level $CL = 1 - \alpha$ for the signal rate can be defined as the set of all values of s (cross section) that are not ruled out by the statistical test. The upper limit, s_{up}, is the maximum value of s that remains consistent with the data. By design, the interval $[0, s_{up}]$ contains the true value of s with a probability of at least 95%, irrespective of the actual value of s.

The limitation of the CL_{s+b} method is that it can lead to the exclusion, with a probability close to α, of hypotheses for which the analysis has little or no sensitivity. This situation arises when the expected number of signal events is significantly smaller than the expected background. In such a situation, the distributions of the test statistic q for the b-only and s+b hypotheses are nearly indistinguishable, exhibiting substantial overlap.

To overcome this problem, instead of excluding the s+b model on the basis of p_{s+b} we use

$$CL_s \equiv \frac{p_{s+b}}{1 - p_b} < \alpha \tag{4.7}$$

The p-value is penalized by dividing it by $1 - p_b$. When the distributions $f(q|b)$ and $f(q|s + b)$ are well separated, $1 - p_b$ is only slightly less than one, so the adjustment is minimal and the resulting exclusion is nearly the same as that obtained from the standard p-value p_{s+b}. In contrast, when the sensitivity to the signal model is low, the two distributions lie close to each other, $1 - p_b$ becomes small, and the adjusted p-value is substantially larger. This reduces the risk of excluding signal models in scenarios with poor sensitivity.

4.2 Machine Learning

Machine learning (ML) is a subfield of artificial intelligence (AI) that focuses on the development of algorithms capable of learning patterns from data and making predictions or decisions without being explicitly programmed for every task. Instead of relying on fixed rule-based instructions, ML systems improve their performance through exposure to data, identifying statistical relationships and generalizing them to unseen examples.

Modern machine learning techniques, including deep learning, is rapidly being applied, adapted, and developed for high energy physics. It has brought a lot of new ideas and improvements to HEP in recent years. On the experimental side, this includes data collection, particle reconstruction and selection as well as subsequent analysis. On the theoretical side, ML contributed a lot to better and faster simulation and parameter inference. Overall, ML has not only led to significant improvements of existing algorithms, but also to new ideas for previously considered impossible-to-solve problems.

In this analysis, machine learning techniques are employed to classify events as either signal or background. The classification is carried out using artificial neural networks, a widely used family of supervised learning algorithms. Neural networks are trained on labeled datasets, where the true event categories are known, and they learn to capture complex correlations among the input features. Once trained, the network generalizes this knowledge to unseen data, enabling it to assign new events to the signal or background classes with improved accuracy compared to traditional cut-based methods.

The performance of a machine learning classifier is evaluated by various quality parameters which give a measure of events correctly classified as signal (tp, true positive) or background (tn, true negative) and events incorrectly classified as signal (fp, false positive) or background (fn, false negative). Some of the quality parameters used in this analysis are give below

$$p = \frac{tp}{tp + fp}, \qquad r = \frac{tp}{tp + fn}, \tag{4.8}$$

$$TPR = \frac{tp}{tp + fn}, \qquad FPR = \frac{fp}{fp + tn}, \tag{4.9}$$

Precision (p) and recall (r) are evaluation metrics that depend on the chosen classification threshold, i.e., the probability above which an event is classified as signal. In contrast, the True Positive Rate (TPR) and False Positive Rate (FPR) can be used to construct the Receiver Operating Characteristic (ROC) curve, which provides a threshold-independent measure of the classifier performance.

4.2.1 TabNet Architecture

TabNet is a type of neural network architecture specifically designed for tabular data [27]. Unlike standard multilayer perceptrons (MLPs), which often perform poorly on tabular datasets compared to gradient-boosted decision trees (like XGBoost or LightGBM), TabNet introduces two key innovations:

- Sequential attention mechanism –At each decision step, the network learns to focus selectively on the most relevant features for the prediction task. This mimics how decision trees perform hierarchical splitting but within a differentiable, neural framework.
- Sparse feature selection –Instead of using all input features at every step, TabNet learns sparse masks that determine which features are most important, improving interpretability and efficiency.

Because of this, TabNet combines the representation learning power of deep neural networks with the interpretability and efficiency of tree-based methods, making it well-suited for supervised learning tasks on structured/tabular data.

4.3 Frameworks

Throughout the analysis chain, different frameworks were utilized to efficiently perform each step of the analysis. Frameworks like `Madgraph` and `Pythia` were used for event generation, `TopCPToolkit` was used for ntuple production, `FastFrames` was used for performing object selections and pre-selections and to perform cutflow analysis and `TREx-Fitter` was used for plotting histograms and most importantly for performing the profile likelihood fit to extract expected upper limits.

4.3.1 MadGraph5_aMC@NLO

MadGraph5_aMC@NLO [28] is an event generation framework designed to support phenomenological studies of the Standard Model (SM) and Beyond the Standard Model (BSM). It provides essential tools for calculating cross sections, generating hard-scatter events, and interfacing these events with parton-shower generators for full event simulation. The framework allows simulations at leading-order (LO) accuracy for arbitrary user-defined Lagrangians, and at next-to-leading order (NLO) accuracy for models where such calculations are supported, most notably for QCD and electroweak corrections to SM processes. In addition, MadGraph5_aMC@NLO can compute matrix elements at both tree level and one-loop level, making it a powerful tool for precision theoretical predictions and event generation.

4.3.2 Pythia8

Pythia [29] is another event generator used in this analysis. It simulates collisions involving particles such as electrons, protons, photons, and heavy ions. It incorporates theoretical models to describe a broad range of physical processes, including both hard and soft interactions, parton distribution functions, initial and final-state parton showers, multiparton interactions, as well as hadronization and particle decays. In this analysis MadGraph5_aMC@NLO is used to generate the hard scatter process which is subsequently interfaced with Pythia8 for simulating parton shower.

4.3.3 TopCPToolkit

`TopCPToolkit` [30] is a framework that is used to produce *ntuples* from derivation files. ntuples are basically root files containing information about all the particles produced in the events. The framework is built around common CP and analysis algorithms used by the ATLAS Athena Software, which is adopted from the GAUDI framework of LHCb. It is used to add decorations, define quality of physics objects, apply pre-selections and perform skimming, thinning and slimming to the derivation files.

4.3.4 FastFrames

`FastFrames` [31] is another framework, built in C++ and python interfaced with ROOT, used to process ntuples. It uses configuration files, which defines various settings and input datasets to perform object selections and event selections, allowing the analyser to reduce the size of the ntuples to contain only interesting events neccessary to carry out the analysis efficiently. `FastFrames` can also be used to perform a cutflow analysis and plot basic histograms for variables. It can also be used to define new higher-level variables from the basic ones already defined in the input ntuples.

4.3.5 TREx-Fitter

`TREx-Fitter` stands for Top Related Experiment Fitter [32]. It is also a framework built in C++ and python interfaced with ROOT. It is used to perform binned template profile likelihood fits for statistical analysis of experimental data. It uses RooFit to build statistical models in HistFactory format and uses RooStats to perform interval estimation and hypothesis testing. It uses a profile likelihood ratio as test statistic and uses various procedure like MINOS and MINUIT to perform the fit [25].

4.3.6 ROOT

ROOT is a software framework for data analysis and I/O: a powerful tool to cope with the demanding tasks typical of state of the art scientific data analysis. Among its prominent features are an advanced graphical user interface, ideal for interactive analysis, an interpreter for the C++ programming language, for rapid and efficient prototyping and a persistence mechanism for C++ objects, used also to write every year petabytes of data recorded by the Large Hadron Collider experiments [33].

5 Search for Asymmetric Pair Production of Scalar Leptoquarks in the $2\ell SS1\tau$ Channel

This chapter presents the full analysis workflow for the search for asymmetric pair production of scalar leptoquarks in the $2\ell SS1\tau$ channel. It begins with an overview of the analysis strategy and a description of the datasets used, including both recorded data from ATLAS Run 2 and Monte Carlo (MC) simulation samples for signal and background processes. The reconstruction and selection of physics objects, as well as the procedures for event selection and overlap removal are detailed, followed by the definition of the signal and control regions. Methods for background estimation are introduced, together with the strategy for feature construction and the implementation of a machine learning classifier to enhance signal–background discrimination. Finally, the statistical framework used to extract upper limits on the leptoquark production cross section is described, along with the presentation of the expected results.

5.1 Analysis Strategy

This section gives an overview of the analysis chain. The analysis starts with the production of MC samples using event generators like MADGRAPH5_aMC@NLO [28], PYTHIA8 [34] etc. These samples are produced centrally on the CERN grid, which takes care of event generation, detector simulation, event reconstruction and derivation production. The derivation files[1] corresponding to each process are obtained and processed through `TopCPToolkit` [36] to produce ntuples for further analysis. These ntuples are processed through `FastFrames` [31] by applying object selections and event selections. Some basic pre-selections are applied using `FastFrames` to reduce the size of the dataset for efficient analysis. These ntuples are used to begin the main analysis, which proceeds by reclassifying the background into fake and prompt leptons. Five different classifications of fake leptons are made and four different control regions orthogonal to the signal region are built for determining the agreement between recorded data and MC samples. A TabNet classifier is trained in the signal region to separate signal and background events and the trained model is used to obtain an inference on recorded

[1]Derivation files or DAODs [35] are the starting point for most analyses. They are made either from the output of reconstruction step or event generation step, and are written in the xAOD data structure. This means they can be read directly with the main analysis frameworks, and many of their variables can be read directly by `ROOT`.

data and MC samples. The signal probability is calculated and added as a new feature in all the datasets. The signal region is used to perform a profile likelihood fit on the effective mass and the signal probability to obtain the sensitivity of the analysis pipeline.

5.2 Datasets

All datasets used in this analysis are in the form of `ROOT` files, referred to as *ntuples*. Each ntuple contains a tree structure with multiple branches, where the branches correspond to the different physics variables relevant to the study. The analysis workflow begins with centrally produced derivation files, which are subsequently processed into ntuples using the `TopCPToolkit` framework. This procedure is applied to both recorded collision data and MC simulation samples. For the present analysis, the specific derivation format used is `DAOD_PHYS`[2].

5.2.1 Recorded Data

The recorded data used in this analysis is from the Run 2 data taking period of the LHC collected by the ATLAS detector in the years 2015 to 2018. The data corresponds to proton-proton collisions at $\sqrt{s} = 13\,\text{TeV}$ with an integrated luminosity of $140\,\text{fb}^{-1}$. The data was collected during stable beam conditions and with all detector subsystems operational [38].

5.2.2 Monte Carlo Samples

The first step in the production of MC samples is preparing the control option and job option files, which are used for event generation step carried out centrally on CERN grid. These files describe the process being simulated and additional settings required for the analysis like showering, and filters to select interesting events.

The signal samples were asymmetrically pair produced scalar leptoquark samples generated using MadGraph5_aMC@NLO with `NNPDF3.0NLO` [39] PDF set interfaced with Pythia8 for showering and hadronization with `NNPDF2.3LO` [40] PDF set. The madgraph model used was `LO_LQ_S1_R2_Bmass` [41]. There were 3 different sets of signal samples produced consisting of 28 different mass points. Two sets assumed same mass hypothesis for the produced leptoquark pair and one set assumed different mass hypothesis. For the same mass hypothesis, one set with Yukawa coupling hypothesis, $y = 0.5$ with 11 mass points ranging from 1500-2500 GeV in steps of 100 GeV and a second set with Yukawa coupling hypothesis, $y = 1.0$ with 11 mass points again ranging

[2] DAOD_PHYS is a new analysis format of derivation files introduced for Run 3 analyses [37]

from 1500-2500 GeV in steps of 100$\dot{\text{G}}$eV were produced. For different mass hypothesis a third set with Yukawa coupling, $y = 1.0$ with 6 mass points 1500 and 2500, 2500 and 1500, 1700 and 2300, 2300 and 1700, 1900 and 2100, 2100 and 1900 GeV were produced. All the MC samples produced centrally at CERN are assigned a Data Set Identifier (DSID) which uniquely determines the process being simulated. A full list of DSIDs for all the signal samples is given in appendix A.1.

Similar to signal samples, the background samples used in the analysis were produced centrally on CERN grid. The dominant backgrounds in the analysis are ttH, ttW, $t\bar{t}$, $Z - jets$ and $W - jets$ samples with $t\bar{t}$, $Z - jets$ and $W - jets$ being major sources of fake leptons. A full list of background samples used along with their DSIDs can be found in appendix A.2.

5.3 Object Selection

Physics objects are objects like electrons, muons, taus, jets and missing transverse momentum. These objects are reconstructed by algorithms using information from the actual detector or detector simulation. It is important to ensure that the physics objects being reconstructed are of good quality, as this ensures better sensitivity to events of interest. In this analysis, the object selection is applied in two steps. First, in `TopCPToolkit`, while producing ntuples from `DAOD_PHYS` derivation files, and second in `FastFrames`. Object selections applied in `TopCPToolkit` are summarized in the next few paragraphs.

Electrons are reconstructed from energy deposits in the electromagnetic calorimeter associated with a charged particle track in the inner detector. To select only electrons originating from the primary vertex, the candidates are required to have $|d_0|/\sigma_{d_0} < 10$ and $|\Delta z_0 \sin\theta| < 0.5$ mm, where d_0 and z_0 are transverse and longitudinal impact parameters respectively. For higher purity, the electron candidates are required to pass a tight likelihood working point ID and a loose variable radius isolation criterion. Additionally, the recontructed elctrons are also required to have $p_T > 10$ GeV and pseudorapidity $|\eta| < 2.47$.

Muons are reconstructed using information from tracks in the inner detector and the muon spectrometer. To ensure that muons originate from the primary vertex, they are required to have $d_0/\sigma_{d_0} < 10$ and $|\Delta z_0 \sin\theta| < 0.5$ mm. For high purity, the candidates are required to satisfy a loose working point ID and a loose variable radius isolation criteria. The muons are further required to have $p_T > 10$ GeV and $|\eta| < 2.5$.

Jets are reconstructed using the anti-k_t algorithm with particle flow objects as inputs [42]. They are required to have $p_T > 20\,\text{GeV}$ and $|\eta| < 4.9$. To reduce the number of jets coming from pile-up a neural network jet vertex tagger is used.

The τ lepton candidates are required to have $p_T > 20\,\text{GeV}$ and $|\eta| < 2.5$. In ATLAS, hadronically decaying τ-leptons are identified using multivariate techniques designed to distinguish genuine τ decays from jets misidentified as τ candidates. A commonly used approach is based on a Recurrent Neural Network (RNN), which exploits the sequential structure of calorimeter clusters and tracks associated with a τ candidate. The output of this algorithm is referred to as the RNN score, a continuous variable between 0 and 1, with values closer to 1 corresponding to a higher likelihood that the candidate is a genuine hadronic τ decay. The τ candidates are required to pass a baseline criterion with an RNN score cut at 0.01 with e-veto.

In `FastFrames`, to improve the quality of objects, another set of object selection is applied to each object. Electrons are required to pass tight likelihood working point along with loose variable radius isolation criterion. They are required to have an ambiquity type of 0, which increases the possibility of the object being an electron and they are also required to have $|\eta| < 2.5$. Muons are required to pass a medium working point with loose variable radius isolation criteria. They are also required to have $|\eta| < 2.5$. τ leptons are required to pass a medium RNN working point and Jets are required to have $|\eta| < 2.5$. A full list of object selection applied in `FastFrames` is shown in Table 5.1.

Physics Object	**Selection**
Electron	`el_select_TightLH_LooseVarRad && el_AmbiguityType == 0 && abs(el_eta) < 2.5`
Muon	`mu_select_Medium_NonIso && mu_select_Loose_Loose_VarRad && abs(mu_eta) < 2.5`
Tau	`MediumRNN`
Jets	`abs(jet_eta) < 2.5`

Table 5.1: Object selection applied in fast frames

5.3.1 Overlap Removal

In ATLAS, the various object collections are reconstructed in separate steps using the same detector information. This means that if, e.g., the electron reconstruction algorithm changes, electrons can be re-reconstructed without needing to change how jets are reconstructed. The downside to this approach is that the different object reconstruction algorithms do not take into account what detector signals are used to build other objects. In other words, the same energy deposited in a region of the calorimeters could be reconstructed as an electron or photon, a hadronic jet, and a tau lepton at the same time. This ambiguity in object reconstruction can lead to counting the same energy deposits multiple times.

ATLAS analyses use a procedure of Overlap Removal (OLR or OR) to resolve this issue. The OLR procedure compares two types of objects at a time and if there is geometric overlap (defined using ΔR matching, ghost-association, or association to the same ID track), one object or the other is removed, depending on a pre-defined priority, which is given below:

- Any calorimeter-tagged muon found to share a track with an electron is removed.
- Any electron found to share a track with a muon is removed.
- Any jet found within a ΔR of 0.2 of an electron is removed.
- Any electron subsequently found within ΔR of 0.4 of a jet is removed.
- Any jet with less than three tracks associated to it found within ΔR of 0.2 of a muon is removed.
- Any jet with less than three tracks associated to it which has a muon inner-detector track ghost-associated to it, is removed.
- Any muon subsequently found within ΔR of 0.4 of a jet is removed.
- Any tau found within a ΔR of 0.2 of a LooseLH electron is removed.
- Any tau found within a ΔR of 0.2 of any type of muon with pT greater than 2 GeV is removed, while noting that if the tau pT is greater than 50 GeV, it will only be removed if it is found to overlap with a combined-type muon.
- Any jet found within a ΔR of 0.2 of a tau is removed.
- Any photon found within a ΔR of 0.4 of an electron or a muon is removed.
- Any jet found within ΔR of 0.4 of a photon is removed.

The different working point criteria used for selecting better qualities of physics objects are also responsible for overlap removal.

5.4 Event Selection

Event selection refers to the process of selecting events that meet certain criteria to qualify as interesting events. Event selection is also performed in two steps in this analysis, first in `TopCPToolkit` and then in `FastFrames`. In `TopCPToolkit` the events are required to pass the Good Runs List (GRL) and have one reconstructed primary vertex (PV). GoodCalo events are selected to reduce discrepancies in jet reconstruction due to noise in the calorimeter. Events with jets passing loose working point are selected and events are required to pass at least one of the following selections:

- Zero Lepton: $N_\tau = 1$, $N_{e,\mu} = 0$,
- One Lepton: $N_\tau = 1$, $N_{e,\mu} = 1$,
- Two Lepton: $N_\tau = 1$, $N_{e,\mu} = 2$,
- Zero Tau: $N_\tau = 0$ and at least one $l = e, \mu$ with $P_T^l > 150\,\text{GeV}$ and $MET > 150\,\text{GeV}$.

The events are also required to pass at leas one of the following triggers: Single Muon, Single Electron, Di Muon, Di Electron, Electron Muon, Single Tau, Di Tau, Tau Muon, Tau Electron and MET trigger.

Events are then passed through a set of pre-selections applied in Fast-Frames to further reduce the size of the datasets to manageable levels for efficient analysis. The events are required to have two light leptons and one hadronically decaying τ lepton. The leading letpon and sub-leading lepton in the events are required to have a transverse momentum grater than 25 GeV and 10 GeV, respectively. The events are also required to have at least one jet. Furthermore, the signal region is defined by imposing the condition of same sign on light leptons and requiring the events have at-least four jets out of which at least one should be b-tagged with a 77% working point efficiency. The full list of pre-selection and signal region selection is shown in Table 5.2.

5.5 Fake Leptons

An important challenge in analyses involving multilepton final states is the presence of *fake* or *non-prompt* leptons. Unlike prompt leptons, which are produced directly in the hard scattering or in the decays of electroweak bosons, fake leptons originate from a variety of background processes. These include semi-leptonic decays of heavy-flavour hadrons (such as b and c mesons), photon conversions, hadronic jets misidentified as

Selection	Condition
Pre-Selection	
Number of Light Leptons	2
Number of Tau Leptons	1
Number of jets	≥ 1
Leading Lepton p_t	$> 25\,GeV$
Sub-Leading Lepton p_t	$> 10\,GeV$
Signal Region Selection	
SS Leptons	
Number of Jets	≥ 4
Number of b-Jets	≥ 1

Table 5.2: Full list of pre-selection and SR selection

leptons, and charge misidentified leptons [43].

Fake leptons are a major background in mulitlepton channels like $2\ell SS1\tau$. This final state is particularly sensitive to processes like $t\bar{t}H$, $t\bar{t}W$, and searches for new phenomena, but it also suffers from sizeable contamination from events in which one or more reconstructed leptons do not originate from the primary interaction. The dominant source of such fakes arises from $t\bar{t}$ production, where the decay of b-hadrons produces secondary electrons or muons that can pass the lepton identification criteria [9].

Monte Carlo simulations alone do not provide a sufficiently reliable estimate of these fake lepton contributions, therefore, ATLAS employs *data-driven methods* to determine the fake lepton background. One widely used approach is the **Template Fit method** [44], which is a *semi-data-driven method* where events are categorised according to their origin (e.g. fake electrons, fake muons, fake taus, or combinations) using truth information in simulation. Control regions orthogonal to the signal region are defined to constrain the normalization of each contribution. A simultaneous fit to data in these regions provides the normalization factors, which are then applied to correct the Monte Carlo estimates in the signal region.

5.6 Background Determination

The backgrounds in the analysis are estimated from the MC samples with the aid of control regions (CR). The dominant backgrounds in the analysis are ttH, ttW, $t\bar{t}$, $Z-jets$ and $W-jets$ samples with $t\bar{t}$, $Z-jets$ and $W-jets$ being major sources of fake leptons. To estimate the fake lepton contributions the MC samples were reclassified into prompt and five different categories of fake leptons using the Isolation and Fake Forum (IFF) tool of ATLAS Athena software which utilises the truth information from MC samples to classify them on the basis of their source and type. The five different categories of fake leptons were as follows:

- Fake e: Events containing exactly one fake electron.
- Fake μ: Events containing exactly one fake muon.
- Fake τ: Events containing exactly one fake τ lepton.
- Double Fake: Events with both light leptons classified as fake.
- Fake $l+\tau$: Events with fake τ lepton and at least one fake light lepton.

A full list of conditions used to re-classify the events is given in appendix A.3. These re-classifications are mutually exclusive and one event can only contribute to one of these re-classifications. Table 5.3 shows event yields from all the backgrounds in the signal region. From the yield table it is evident that the major contribution in the SR comes from Fake e, Fake μ, Fake τ and Fake $l+\tau$. Double Fake samples were regrouped with "others" because of lower statistics to ensure stability of background modeling. To estimate major fake contributions, four different control regions were prepared orthogonal to the signal region and template fit method was used to obtain the best fit norm factors for these contributions. The control regions were made using the events passing the pre-selections described in Section 5.4 and are made in such a way so that they are mostly populated by one of the aforementioned fake samples, this is done to reduce correlation between the norm factors. Table 5.4 shows the selections used to define the control regions. Figure 5.1 shows the distribution of the effective mass in the signal region along with the weighted event yields mentioned for each MC sample and recorded data. There was a blinding threshold of 0.1 defined to ensure that the recorded data are not visible in the signal region and the analysis is not biased.

Sample	SR Yield (Weighted)
Fake e	58.672 ± 0.862
Fake μ	85.196 ± 1.252
Fake τ	43.413 ± 0.636
Fake $l+\tau$	226.842 ± 3.344
ttH	31.689 ± 0.464
Double Fake	3.580 ± 0.054
Others	34.374 ± 0.484

Table 5.3: Pre-fit event yields in the signal region (SR)

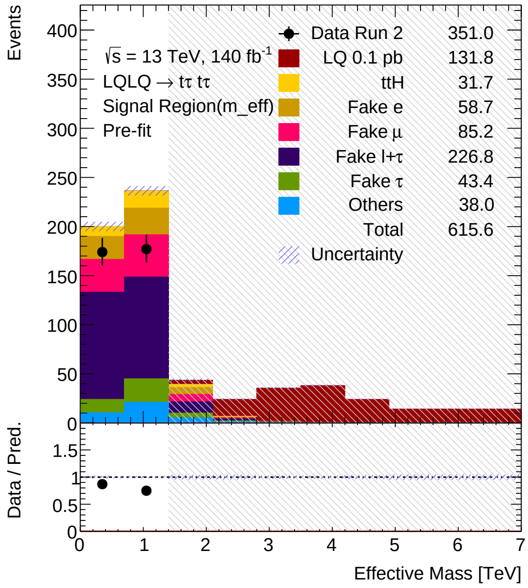
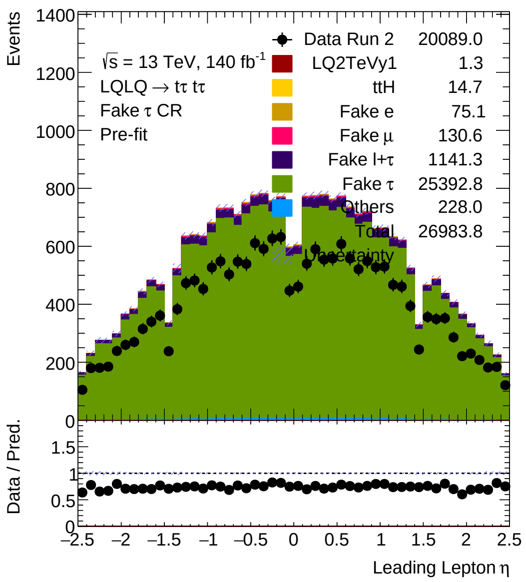


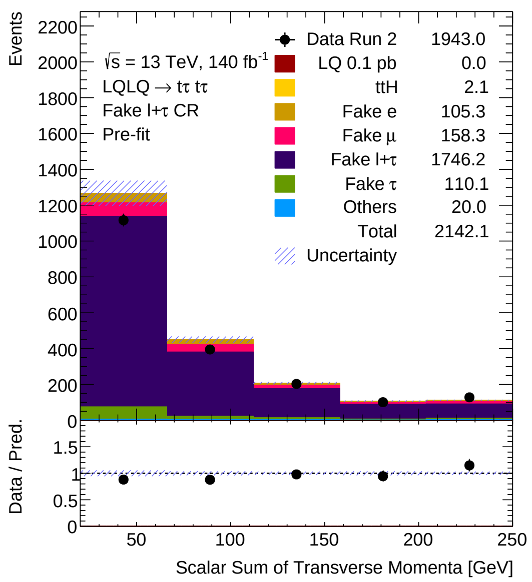
Figure 5.1: Distribution of effective mass in the signal region for signal and all the background samples. The dominant background comes from fake leptons. A blinding threshold of 0.1 has been applied to avoid biasing the analysis.

Fake τ	Fake $l+\tau$	Fake e	Fake μ
$m_{eff} < 1.5\,TeV$	$m_{eff} < 1.5\,TeV$	$m_{eff} < 1.2\,TeV$	$m_{eff} < 1.5\,TeV$
Number of Jets ≤ 3	Number of Jets ≤ 3	Number of Jets ≤ 3	Number of Jets ≤ 3
OS Leptons	SS Leptons	SS Leptons	SS Leptons
Leading Lepton $p_T > 100\,GeV$	Tau $p_T < 40\,GeV$	Tau $p_T > 35\,GeV$	Tau $p_T > 40\,GeV$
-	Number of b-Jets $= 0$	Number of b-Jets ≥ 1	Number of b-Jets ≥ 1
-	-	tau_select_TightRNN	tau_select_TightRNN
-	-	Only Reconstructed Electrons	Only Reconstructed Muons

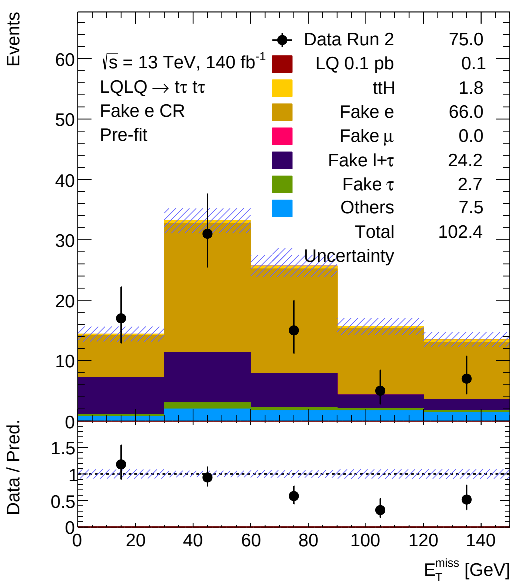
Table 5.4: Overview of selection in the CRs. The control regions have been constructed such that they are mostly populated by the corresponding fake background.



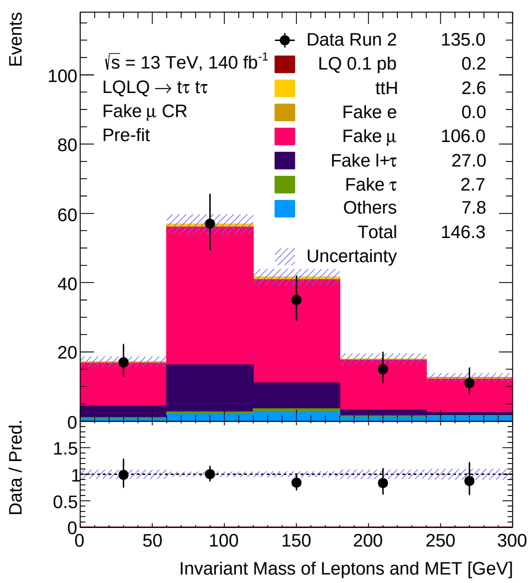
(a) Fake τ CR



(b) Fake $l + \tau$ CR



(c) Fake e CR



(d) Fake μ CR

Figure 5.2: Control region plots for $m_{LQ} = 2.0\,\text{TeV}$, same mass hypothesis.

5.7 Feature Addition

The ntuples initially produced by `TopCPToolkit` contain very basic variables (features), `FastFrames` allows the addition of more features which can be added while processing the ntuples to apply pre-selections. In any BSM physics analysis targeting signal and background separation, it is important that the variables used in the analysis have distributions significantly different for signal and background samples. Such variables are called high-level variables and are crucial for the analysis. These features can not only be used to give good separation in a cut-based analysis but can also prove extremely important for training a machine learning classifier. There were several high-level variables added in the analysis which were further used to train a machine learning model for signal-background separation. One of the most important high-level variables added in the analysis was the effective mass, which is the invariant mass of all the final state objects. In the signal region this corresponds to the invariant mass of two light leptons, one tau lepton, all the jets, and the missing transverese energy. Figure 5.3[3] shows the fifteen most important features used for training and Figure 5.4 shows the correlations of the fifteen most important features. Features with high correlations to other features were removed to ensure that the features used for training contain new information. A full list of features used for training the classifier is given in appendix A.4 and A.5. Appendix A.6 also shows distributions of ten most important features.

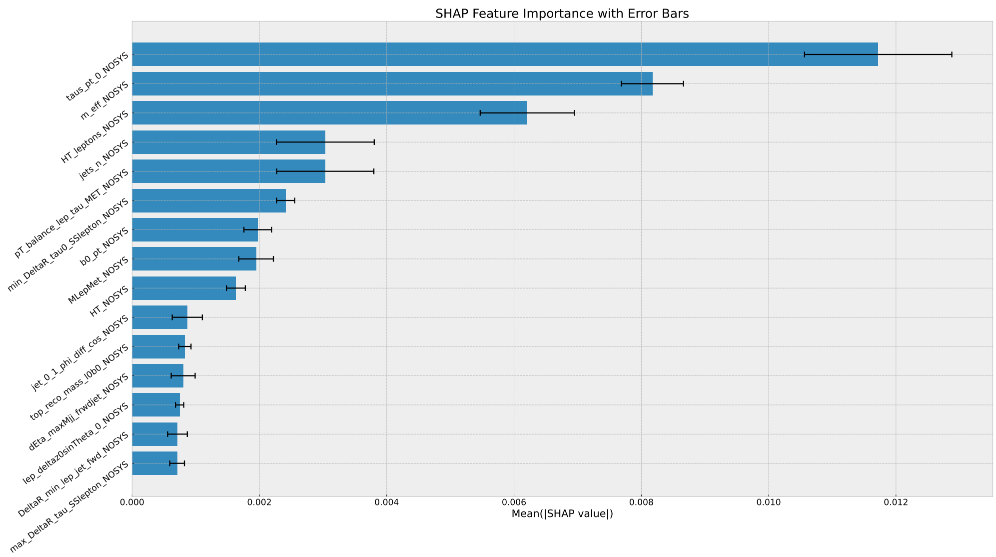


Figure 5.3: 15 most important features for training

[3]The suffix “$NOSYS$” means that the effect of systematics has not been accounted for in the analysis

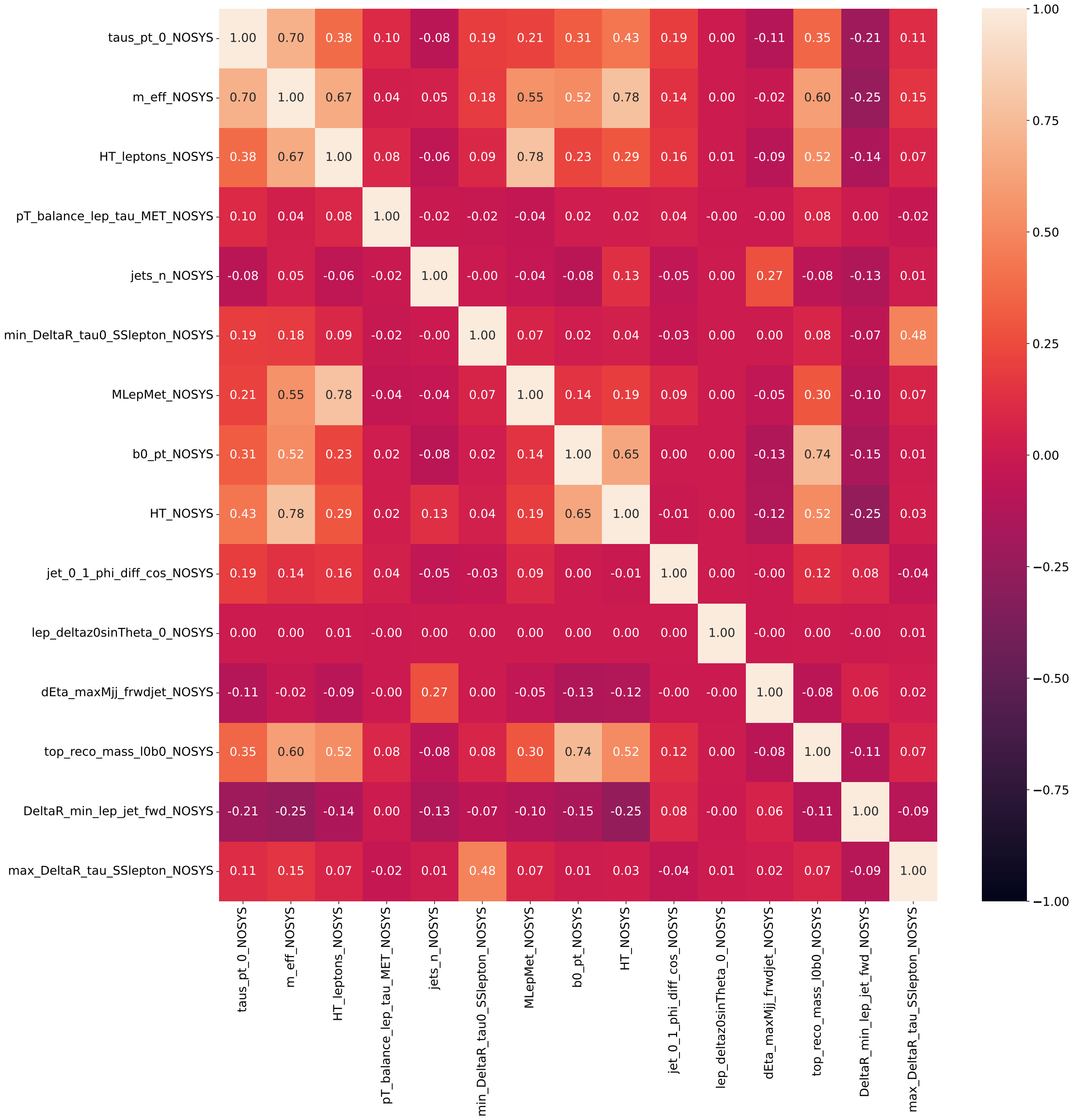


Figure 5.4: Correlations of most important features

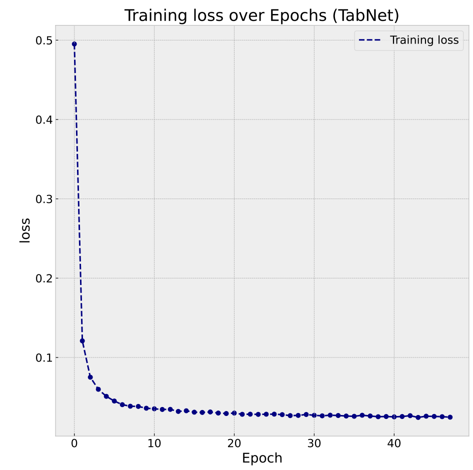
5.8 Machine Learning

For the classification task in this analysis, a **TabNet** model was implemented using `PyTorch`. The model was configured with $n_d = 32$ decision dimensions and $n_a = 32$ attention dimensions across $n_{\text{steps}} = 7$ decision steps, and a relaxation parameter $\gamma = 1.1$.

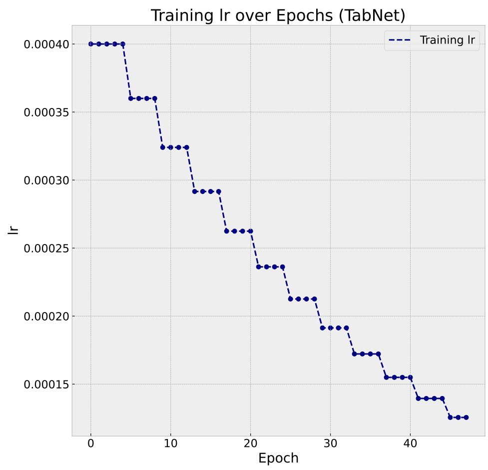
To encourage sparsity in the feature selection, a regularization strength of $\lambda_{\text{sparse}} = 10^{-5}$ was applied, together with a momentum parameter of 0.05. The optimization was performed using the AdamW optimizer with a learning rate of 4×10^{-4} and weight decay of 10^{-4}. A learning rate scheduler based on the `ReduceLROnPlateau` strategy was used, with mode set to "min" (optimizing the validation loss), patience of 3 epochs, and a minimum learning rate of 10^{-5}. The loss function used was the cross-entropy loss, weighted by per-sample training weights to account for imbalances in the dataset. Model performance was monitored using both classification accuracy and the area under the ROC curve (AUC) as evaluation metrics. Training was performed in the signal region which was split into training and validation sets and the model was initially configured to run for 50 epochs. An early stopping criterion with a patience of 8 epochs was applied in order to prevent overfitting and reduce unnecessary computation. As a result, training converged early and terminated after 47 epochs, at which point no further improvement in the validation performance was observed. Figure 5.5 shows the performance of the model on both the training and the validation sets.

A consistent drop in the loss function and the learning rate is seen for the training set, indicating that the classifier trained accurately and actively adapted to the training process. The staircase structure of the learning rate demonstrates the fact that the `ReduceLROnPlateau` scheduler works as expected, decreasing the learning rate if significant improvement in the performance is not observed for three consecutive epochs as defined in the intial setting of patience=3. This stabilizes the classifier performance and refines the convergence. A consistent increase in accuracy and area under the ROC curve(AUC) is observed for the validation set, indicating that the classifier is good at generalizing and shows no signs of overtraining. Both metrics demonstrate that the model learns effectively in the initial training phase, the plateau indicates that the model converged quickly and maintains consistently high classification accuracy. The near-perfect scores demonstrate strong discriminative power and suggest that TabNet captures the underlying differences between the signal and background with very few misclassifications. Figure 5.6 also shows the ROC curve plotted for the TabNet model with the optimal threshold point marked red. This point has been calculated by maximizing the Youden's J score which is the difference between TPR and FPR. It is observed that the classifier has an $AUC = 0.9995 \approx 1$ again signifying its accurate performance irrespective of the classification threshold.

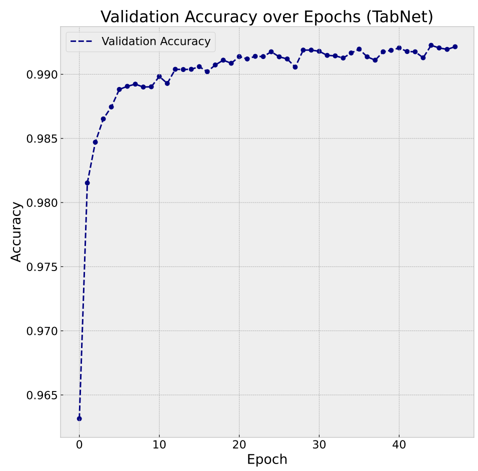
The trained model was used to perform inference on all the MC simulation samples as well as the recorded data in the signal region. It assigns to each event the probability of being a signal event with close to 1 being high chances of the event being a signal event and close to 0 being low chances of the event being a signal event. The calculated probabilities were appended to the ntuples as a new feature using uproot and precision and recall was calculated for various values of threshold. Figure 5.7 shows the dependence of precision and recall scores on the classification threshold for the TabNet model. As expected, increasing the threshold improves the precision at the expense of recall. The intersection of the curves occurs at a threshold of about 0.5, which was adopted as the working point in this analysis to balance the trade-off between the two scores.



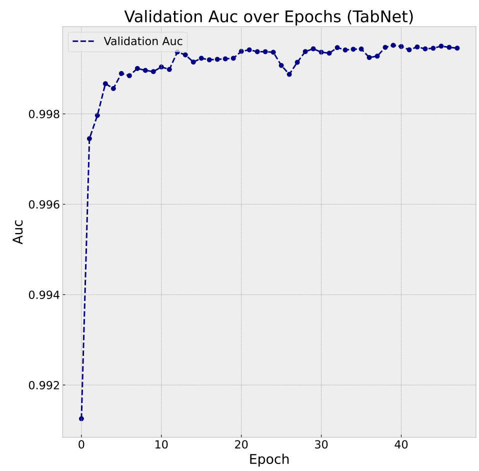
(a) Loss function for training set



(b) Learning rate for training set



(c) Accuracy for validation set



(d) Area under the ROC curve for validation set

Figure 5.5: Performance for TabNet model

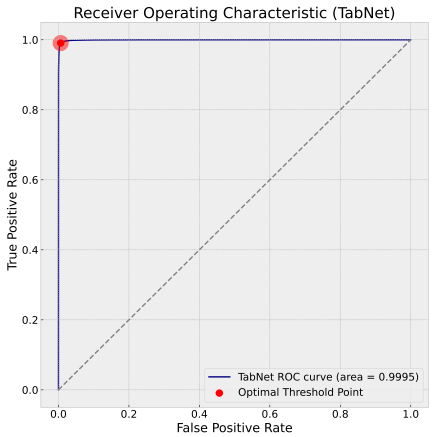


Figure 5.6: ROC curve

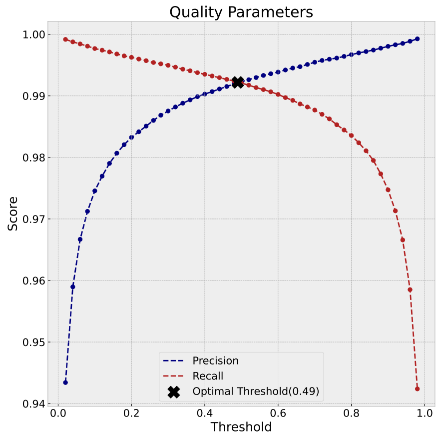


Figure 5.7: Quality parameters as a function of threshold

Figure 5.8 shows the confusion matrix obtained by using the optimal threshold as the classifying threshold. Thus, the model classifies 99.20% events correctly from the full pool of all the MC events. This does not contain events from recorded data as true labels for recorded data do not exist. Figure 5.9 shows the distribution of the calculated signal probability in the signal region.

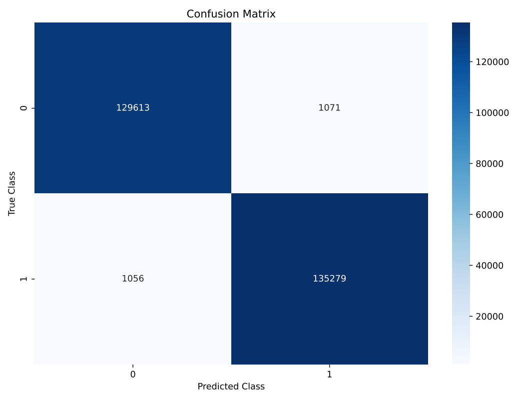


Figure 5.8: Confusion matrix for optimal threshold

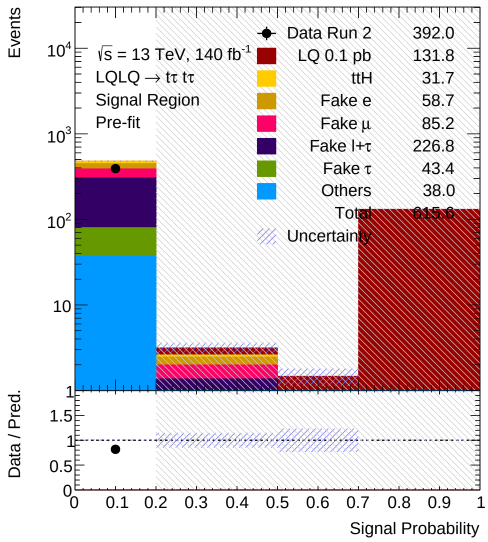
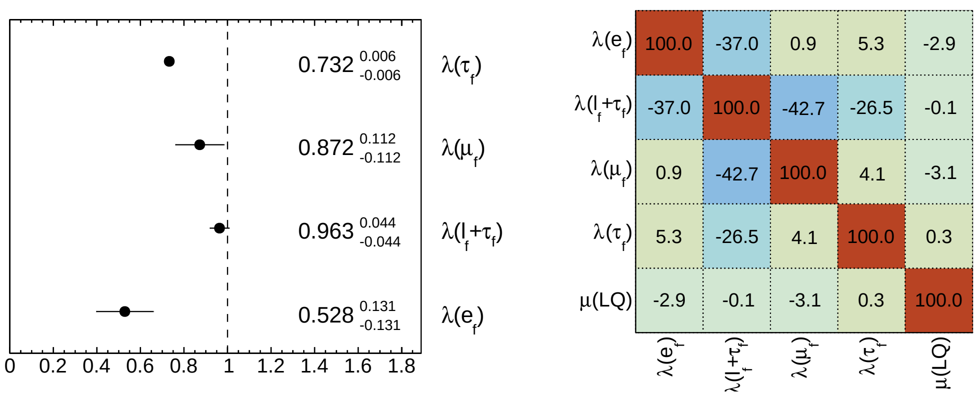


Figure 5.9: Signal probability distribution in SR

5.9 Results

The background only hypothesis and the signal+background hypothesis was defined using MC samples and the signal+background hypothesis was defined separately for all the 28 mass points. Norm factors for major fake contributions were defined and the best fit values of these norm factors were obtained using the template fit method. This was done by performing a simultaneous CR+SR fit to the data. Figure 5.10a shows the best fit values of these norm factors and Figure 5.10b shows the correlations between the norm factors. These plots have been shown for signal hypothesis of $m_{LQ} = 2.0\,\text{TeV}$ and $y = 1.0$ but similar results were obtained for all the signal samples indicating stability of the background modeling. The value of $\lambda(l_f + \tau_f)$ has slightly high correlations with other norm factors, this was because of significant $fake\; l + \tau$ contribution in other control regions. This correlation could be reduced but at the expense of losing statistics. A blinding threshold of 0.1 was used and the best fit value of the POI (i.e. the signal strength, $\mu(LQ)$) was also blinded to ensure that the analysis does not get biased.

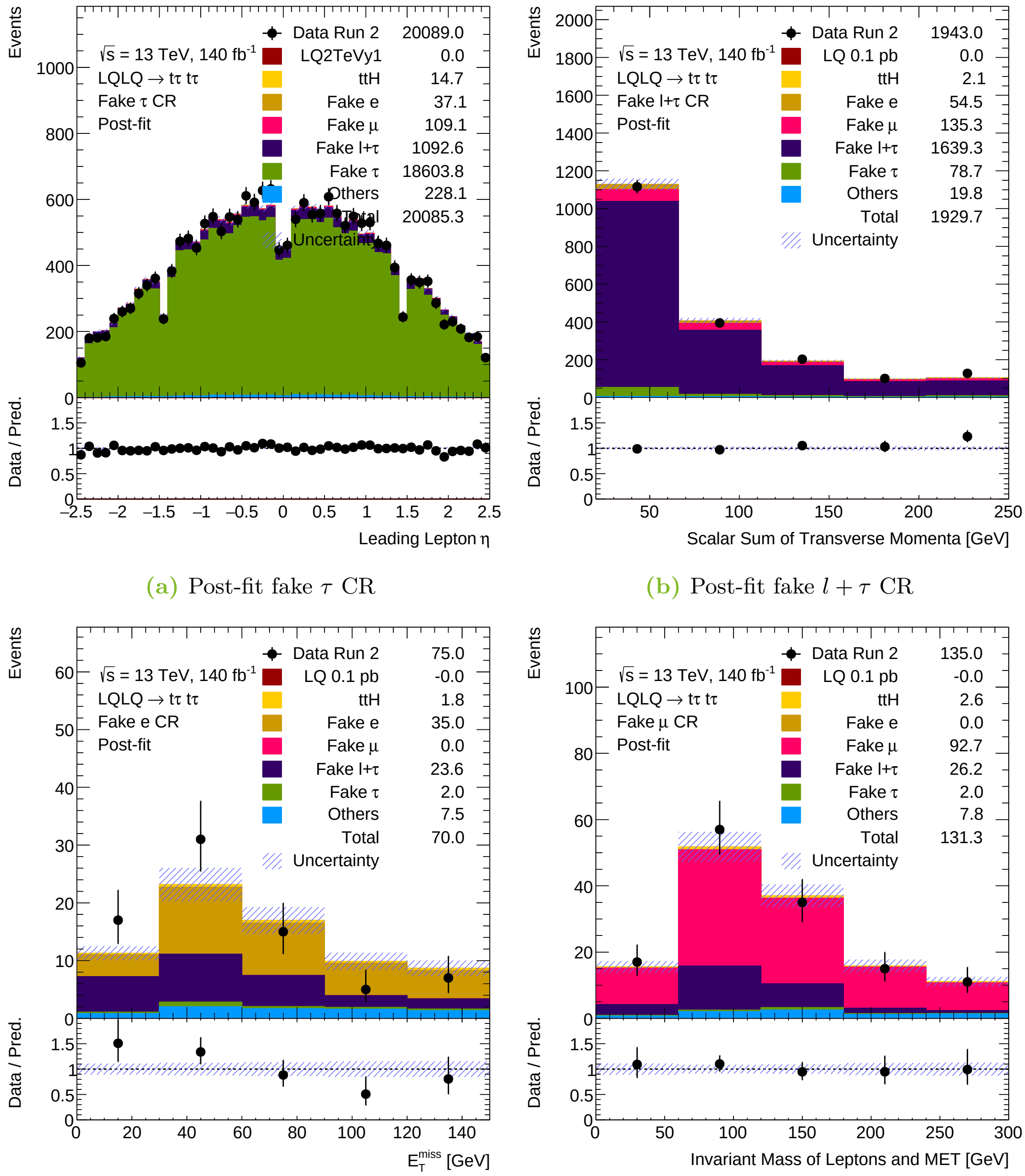


(a) Best fit norm factors

(b) Correlations of norm factors

Figure 5.10: Best fit values and correlations of norm factors. The best fit value of the POI is blinded and hence has not been shown. The correlations of POI to other norm factors is close to zero because of very low signal events in the control regions.

Figure 5.11 shows post-fit plots of the control regions. After fitting, the MC samples show very good agreement with data in the $fake\;\tau$ and $fake\; l+\tau$ regions. This indicates that $fake\;\tau$ and $fake\; l + \tau$ samples have been properly modeled. The $fake\;\mu$ region shows good agreement of MC samples with the recorded data with only a slight deviation in the second bin. The $fake\; e$ region shows slightly more deviations which might be attributed to very low statistics in this region. Apart from the slight discrepancies in some individual bins, the agreement between the total number of events in all the control regions after fitting is adequate, indicating a good modeling of the background.



(a) Post-fit fake τ CR

(b) Post-fit fake $l + \tau$ CR

(c) Post-fit fake e CR

(d) Post-fit fake μ CR

Figure 5.11: Post-fit control region plots for $m_{LQ} = 2.0\,\text{TeV}$, same mass hypothesis.

The model was also fitted to the Asimov dataset produced assuming the background-only hypothesis and was tested against the S+B hypothesis in the signal region to extract the expected upper limits on the cross section. Two different fit variables were used for this purpose—the effective mass and the signal probability. Expected upper limits were extracted for all the mass points for both of these variables separately and compared.

Tables 5.5, 5.6 and 5.7 show the expected 95% CL upper limits on the cross section for all the three sets of samples along with the statistical uncertainties obtained by using effective mass as the fit variable. The tables show that the expected upper limits calculated for $y = 0.5$ and $y = 1.0$ for the same mass hypothesis do not show much difference and are within the uncertainty limits of each other, this hints to the fact that the kinematics of the signal samples do not depend upon the value of the Yukawa couplings, thus, similar sensitivity is expected for different values of the Yukawa couplings. This can be verified by comparing the truth-level[4] kinematic distributions of different particles using the MC simulations. Figures 5.14 and 5.15 compare different kinematic distributions of truth-level particles produced in the process and it can be concluded that the distributions are similar for $y = 0.5$ and $y = 1.0$ hypotheses. Figure 5.12 shows the expected 95% CL upper limit along with 1σ and 2σ error bands. It also shows a theory curve for Yukawa coupling hypothesis $y = 2.0$ overlaid. Since it was concluded that expected upper limits are independent of Yukawa couplings, this plot can be used to establish an expected exclusion of leptoquark masses below about 1980 GeV assuming $y = 2.0$. A similar plot of different mass hypothesis is also shown in Figure 5.13 along with the theory curve for $y = 2.0$. The sensitivity obtained using the effective mass was not enough in the different mass hypothesis to establish an expected exclusion limit for y=2.0.

LQ Mass [GeV]	**Exp. Upper Limit [pb]**	**+1σ**	**-1σ**	**+2σ**	**-2σ**
1500	0.00343	0.00550	0.00219	0.00853	0.00149
1600	0.00317	0.00515	0.00201	0.00807	0.00136
1700	0.00308	0.00506	0.00192	0.00802	0.00128
1800	0.00292	0.00486	0.00180	0.00779	0.00119
1900	0.00302	0.00507	0.00184	0.00818	0.00121
2000	0.00289	0.00491	0.00175	0.00799	0.00114
2100	0.00288	0.00495	0.00172	0.00814	0.00111
2200	0.00277	0.00480	0.00164	0.00795	0.00104
2300	0.00281	0.00491	0.00165	0.00819	0.00104
2400	0.00280	0.00494	0.00163	0.00828	0.00102
2500	0.00280	0.00497	0.00162	0.00837	0.00100

Table 5.5: Expected upper limit table from effective mass (y=0.5)

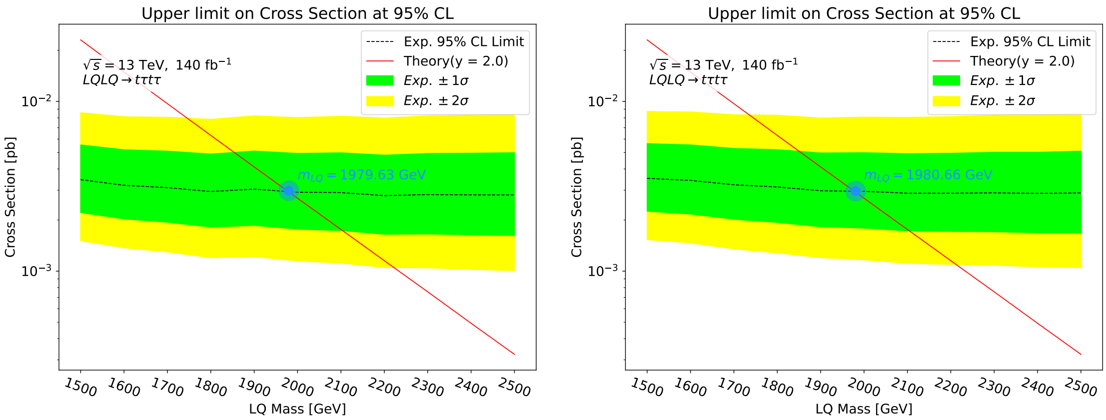
[4] Truth-level particles are particles produced after the event generation step as opposed to reconstructed objects after detector simulation.

LQ Mass [GeV]	Exp. Upper Limit [pb]	$+1\sigma$	-1σ	$+2\sigma$	-2σ
1500	0.00349	0.00560	0.00223	0.00867	0.00152
1600	0.00339	0.00550	0.00215	0.00861	0.00145
1700	0.00320	0.00524	0.00200	0.00829	0.00134
1800	0.00311	0.00516	0.00192	0.00825	0.00127
1900	0.00295	0.00494	0.00181	0.00793	0.00119
2000	0.00293	0.00496	0.00178	0.00804	0.00116
2100	0.00286	0.00489	0.00171	0.00801	0.00110
2200	0.00286	0.00492	0.00170	0.00810	0.00109
2300	0.00288	0.00499	0.00170	0.00827	0.00108
2400	0.00286	0.00499	0.00167	0.00832	0.00105
2500	0.00288	0.00507	0.00167	0.00850	0.00104

Table 5.6: Expected upper limit table from effective mass (y=1.0)

LQ Mass [GeV]	Exp. Upper Limit [pb]	$+1\sigma$	-1σ	$+2\sigma$	-2σ
1500 & 2500	0.00293	0.00495	0.00178	0.00802	0.00116
1700 & 2300	0.00290	0.00491	0.00176	0.00796	0.00114
1900 & 2100	0.00293	0.00495	0.00178	0.00804	0.00115
2100 & 1900	0.00303	0.00512	0.00183	0.00832	0.00119
2300 & 1700	0.00302	0.00513	0.00182	0.00835	0.00118
2500 & 1500	0.00297	0.00505	0.00180	0.00821	0.00117

Table 5.7: Expected upper limit table from effective mass (different mass hypothesis)



(a) Expected upper limits (y=0.5)

(b) Expected upper limits (y=1.0)

Figure 5.12: Expected upper limits from fit to effective mass (same mass hypotheses)

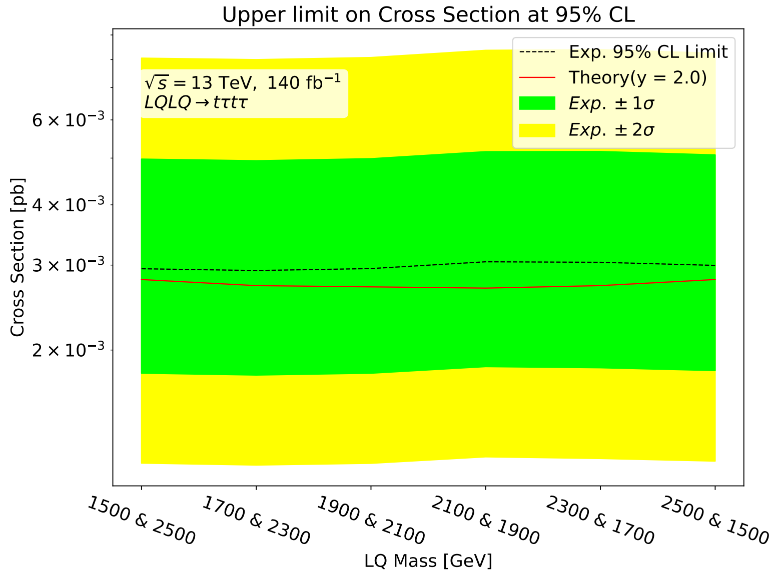


Figure 5.13: Expected upper limits calculated by using effective mass as the fit variable for different mass hypothesis. The calculated expected limit curve lies entirely on top of the theory curve and hence cannot be used to establish an expected exclusion on leptoquark mass.

Tables 5.8, 5.9 and 5.10 show the expected 95% CL upper limits on cross section for all three sets of samples along with the statistical uncertainties obtained using signal probability as the fit variable. The tables show that, for the same-mass hypothesis, the limits obtained using the signal probability are slightly stronger than those derived from the effective mass at lower masses. For the different-mass hypothesis, the limits from the signal probability are consistently better across the entire mass range, improving the sensitivity by a factor of about 0.6. The expected upper limits calculated for $y = 0.5$ and $y = 1.0$ for the same mass hypothesis are within the uncertainty limits of each other in this case as well, verifying that the kinematics of the signal samples is independent of the Yukawa coupling. Figure 5.16 shows the expected 95% CL upper limit along with 1σ and 2σ error bands obtained from fitting on signal probability. The theory curve for the Yukawa coupling hypothesis $y = 2.0$ cuts the calculated curve at approximately the same point as in Figure 5.12, leptoquark masses less than about 2010 GeV are expected to be excluded for $y = 2.0$. Figure 5.17 shows the expected limit plot for different mass hypothesis. In this case the expected limit curve lies entirely below the theory curve and hence the entire range of masses from $1500 - 2500$ GeV is expected to be excluded at 95% CL.

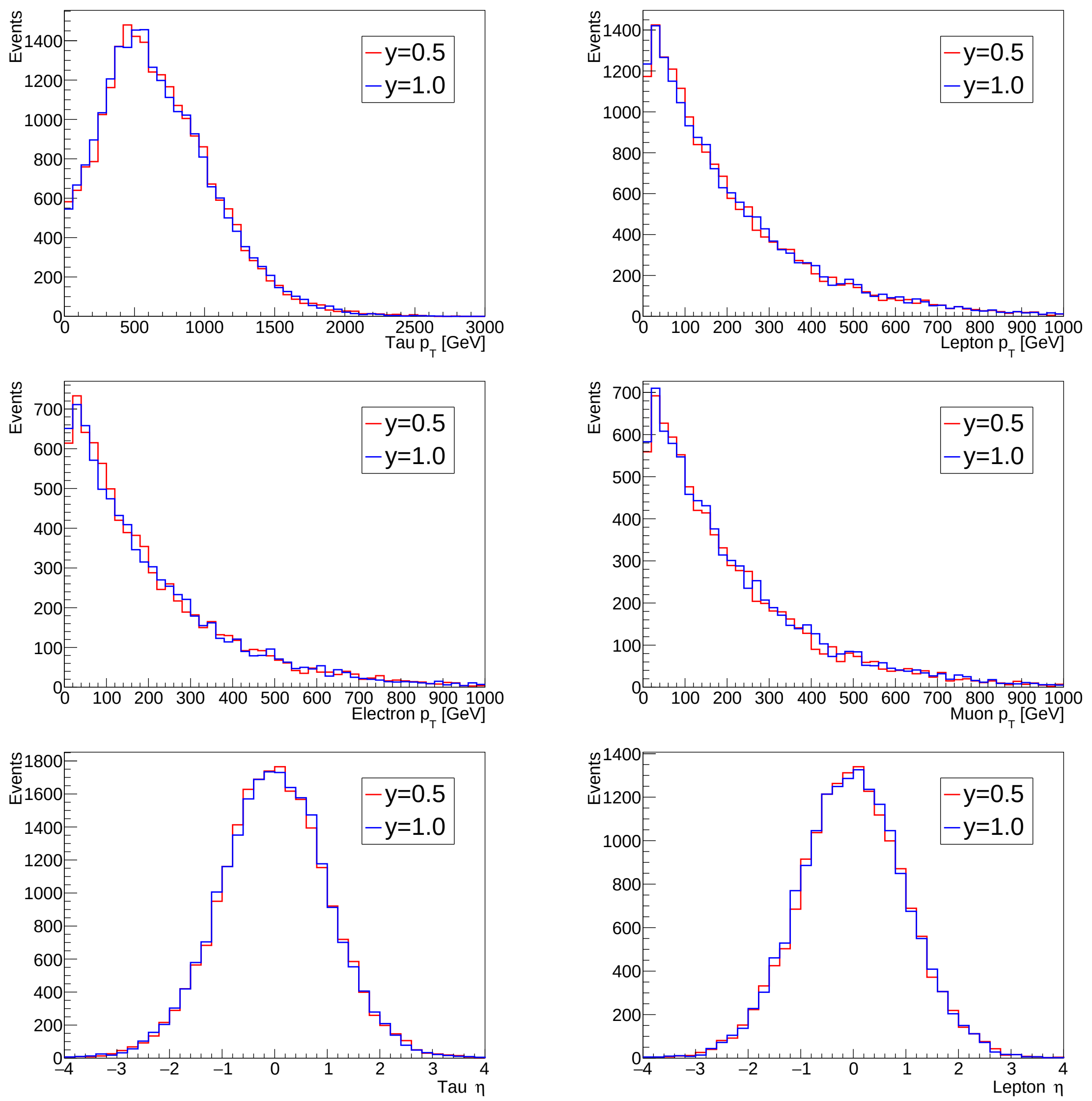


Figure 5.14: p_T and η distributions of truth-level particles for signal sample $m_{LQ} = 1500\,\text{GeV}$ same mass hypothesis.

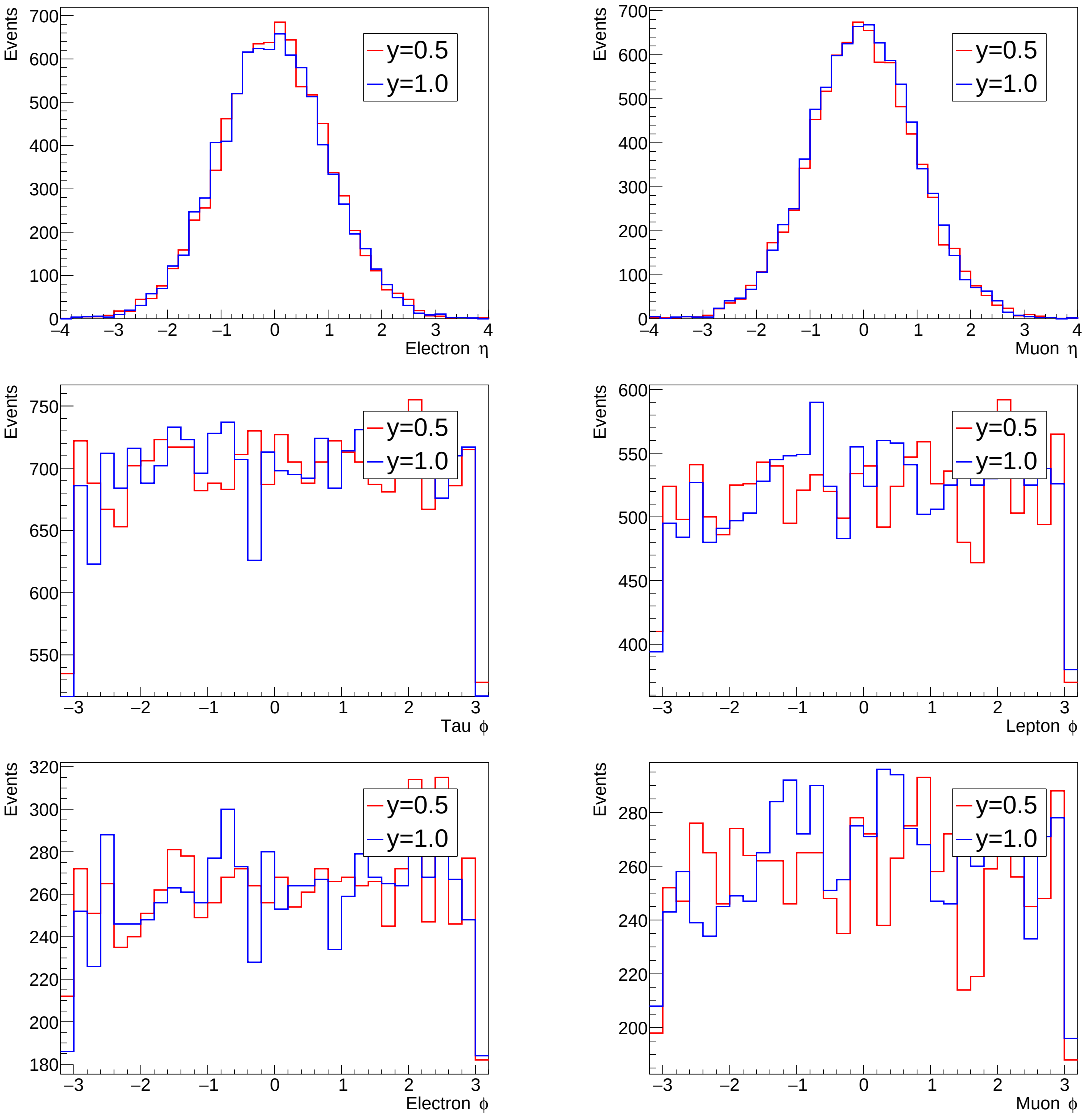


Figure 5.15: η and ϕ distributions of truth-level particles for signal sample $m_{LQ} = 1500\,\mathrm{GeV}$ same mass hypothesis.

LQ Mass [GeV]	Exp. Upper Limit [pb]	$+1\sigma$	-1σ	$+2\sigma$	-2σ
1500	0.00210	0.00350	0.00131	0.00565	0.00088
1600	0.00210	0.00349	0.00131	0.00565	0.00088
1700	0.00222	0.00370	0.00138	0.00598	0.00093
1800	0.00227	0.00379	0.00142	0.00612	0.00095
1900	0.00248	0.00413	0.00155	0.00669	0.00104
2000	0.00253	0.00422	0.00158	0.00682	0.00106
2100	0.00269	0.00447	0.00167	0.00723	0.00112
2200	0.00271	0.00452	0.00169	0.00731	0.00114
2300	0.00286	0.00475	0.00178	0.00769	0.00119
2400	0.00299	0.00498	0.00186	0.00805	0.00125
2500	0.00310	0.00516	0.00193	0.00835	0.00130

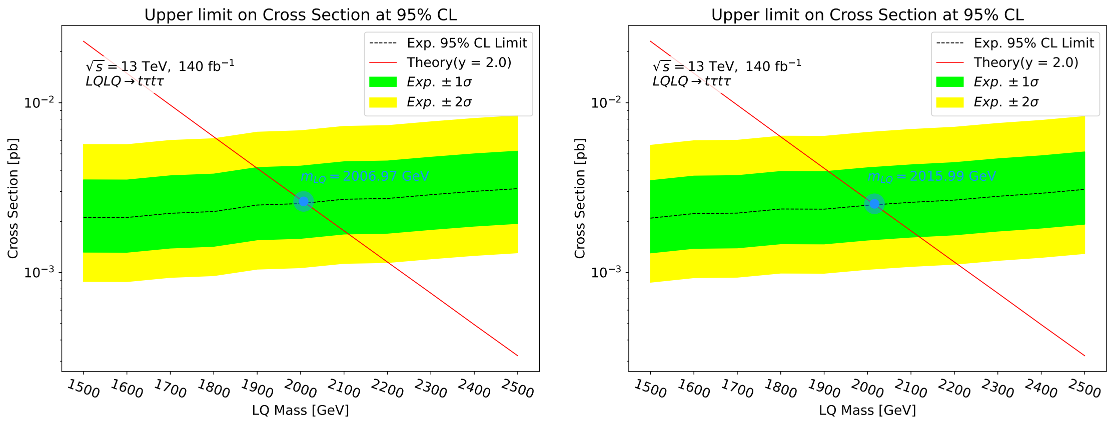
Table 5.8: Expected upper limit table from signal probability (y=0.5)

LQ Mass [GeV]	Exp. Upper Limit [pb]	$+1\sigma$	-1σ	$+2\sigma$	-2σ
1500	0.00208	0.00346	0.00130	0.00560	0.00087
1600	0.00221	0.00368	0.00138	0.00595	0.00093
1700	0.00223	0.00371	0.00139	0.00599	0.00093
1800	0.00235	0.00392	0.00147	0.00634	0.00098
1900	0.00235	0.00391	0.00146	0.00632	0.00098
2000	0.00248	0.00412	0.00154	0.00667	0.00104
2100	0.00258	0.00429	0.00161	0.00694	0.00108
2200	0.00266	0.00442	0.00166	0.00715	0.00111
2300	0.00280	0.00465	0.00174	0.00753	0.00117
2400	0.00291	0.00485	0.00182	0.00784	0.00122
2500	0.00307	0.00511	0.00191	0.00827	0.00128

Table 5.9: Expected upper limit table from signal probability (y=1.0)

LQ Mass [GeV]	Exp. Upper Limit [pb]	$+1\sigma$	-1σ	$+2\sigma$	-2σ
1500 & 2500	0.00245	0.00409	0.00153	0.00661	0.00103
1700 & 2300	0.00245	0.00408	0.00153	0.00660	0.00102
1900 & 2100	0.00248	0.00413	0.00154	0.00667	0.00104
2100 & 1900	0.00256	0.00427	0.00160	0.00690	0.00107
2300 & 1700	0.00261	0.00434	0.00162	0.00701	0.00109
2500 & 1500	0.00257	0.00428	0.00160	0.00692	0.00108

Table 5.10: Expected upper limit table from signal probability (different mass hypothesis)



(a) Expected upper limits (y=0.5) **(b)** Expected upper limits (y=1.0)

Figure 5.16: Expected upper limits calculated using signal probability as fit variable for same mass hypotheses. Fitting on signal probability shows better result for lower masses compared to the fit on effective mass. Leptoquark masses below about 2000 GeV are expected to be excluded.

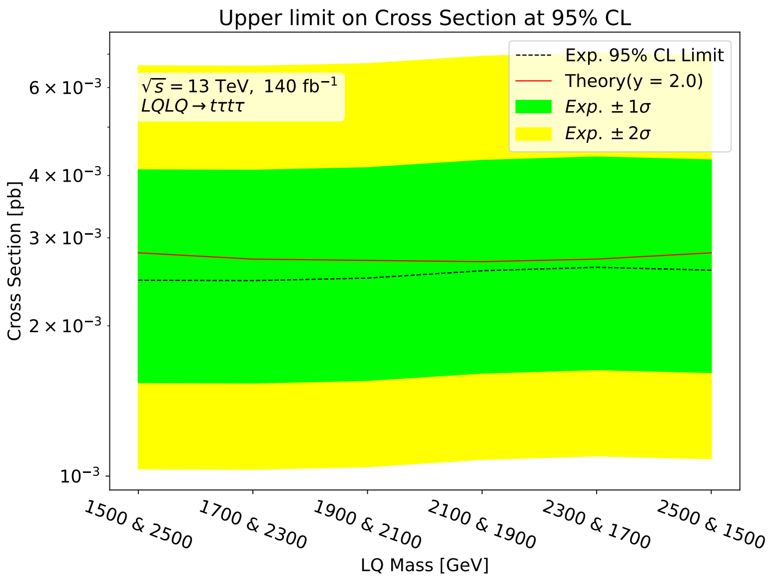


Figure 5.17: Expected upper limits calculated using signal probability as the fit variable for different mass hypothesis. The calculated expected curve lies entirely below the theory curve allowing for the expected exclusion of all the masses in the range 1500 − 2500 GeV.

6 Conclusion and Outlook

This thesis presented a blinded analysis of asymmetric pair production of scalar leptoquarks in the $2\ell SS1\tau$ channel in Release 22 of the ATLAS software. Machine Learning techniques were employed for sensitivity analysis. Monte Carlo signal samples were produced on the CERN grid for 3 different sets—same mass hypothesis (y=0.5), same mass hypothesis (y=1.0) and different mass hypothesis (y=1.0), in total comprising of 28 different signal MC samples. Pre-produced MC samples were used for modeling the background. An extensive object selection and event selection algorithm was employed using `TopCPToolKit` and `FastFrames` to select events of interest. The events were divided into prompt and fake lepton contributions using `TREx-Fitter`. To model the major fake contributions dedicated control regions were made and best fit values of norm factors for fakes were obtained using template fit method. The best fit values of norm factors obtained for different signal hypotheses were observed to be similar, indicating stability in modeling the background. Good agreement was obtained between MC samples and recorded data in the post-fit control regions. A TabNet model was trained using pytorch in the signal region to separate signal and background. The training process showed good results across both the training and the validation sets and the model obtained an $AUC = 0.9995$ indicating a high separating power. The model was used to perform inference on all the MC samples and recorded data in the signal region to get signal probability. Finally, profile likelihood fits were performed in the signal region on the effective mass and the signal probability, and expected 95% CL upper limits on the cross section were obtained. It was established that the kinematics of the signal, and hence the expected upper limits, are independent of the Yukawa couplings of leptoquarks. The fit to the signal probability resulted in better sensitivity and leptoquark masses below about 2000 GeV are expected to be excluded for the same mass hypothesis $y = 2.0$, while all the masses in the range $1500 - 2500$ GeV are expected to be excluded for different mass hypothesis $y = 2.0$.

This thesis only presents an expected exclusion on the leptoquark mass. An actual mass exclusion can only be inferred after a full unblinded analysis is performed in the signal region and the observed cross section of the real data is also calculated and compared with the expected 95% CL upper limit. To further improve the expected sensitivity, systematic uncertainties need to be included. An extension of the mass range can also be considered by including the signal samples from $500 - 1400$ GeV. These suggestions are left to future analyses.

A Appendix

A.1 DSIDs for Signal Samples

DSID	Mass [GeV]	Yukawa Coupling
567820	1500	0.5
567821	1600	0.5
567822	1700	0.5
567823	1800	0.5
567824	1900	0.5
567825	2000	0.5
567826	2100	0.5
567827	2200	0.5
567828	2300	0.5
567829	2400	0.5
567830	2500	0.5

Table A.1: LQ samples (y=0.5)

DSID	Mass [GeV]	Yukawa Coupling
545824	1500	1.0
545825	2000	1.0
545826	2500	1.0
567831	1600	1.0
567832	1700	1.0
567833	1800	1.0
567834	1900	1.0
567835	2100	1.0
567836	2200	1.0
567837	2300	1.0
567838	2400	1.0

Table A.2: LQ samples (y = 1.0)

DSID	Mass [GeV]	Yukawa Coupling
567839	1500-2500	1.0
567840	1700-2300	1.0
567841	1900-2100	1.0
567842	2100-1900	1.0
567843	2300-1700	1.0
567844	2500-1500	1.0

Table A.3: LQ samples (different mass)

A.2 DSIDs for Backgrounds

Sample	DSIDs
$t\bar{t}$	410470
ttH	346343, 346344, 346345
ttW	700168
$Z-jets$	700320-700325, 700335-700337
$W-jets$	700338-700349
Single Top	410659, 410644, 410645, 410654, 410655
VV	701000, 701005, 701010, 701015, 701020, 701025, 701030, 701035, 701055, 701085, 701090, 701095, 701100, 701105, 701115, 701120, 701125
VH	346645, 346646
ttZ	504330, 504342, 504334
ttW	701260, 701261, 701262
tZ	410560
VVV	364242-364249
Three Top	304014
Four Top	412043
WtZ	410408
ttWW, ttWZ, ttHH, ttWH, ttZZ	410081, 500463, 500460, 500461, 500462
$tt\gamma$	500800,504554
$V\gamma$	700398-700404

Table A.4: DSIDs used as background

A.3 Fake Lepton Re-classification

Re-Classification	Selection
Prompt Lepton	`(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && (leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && tau_truth_IsHadronicTau`
Double Fake	`!(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && !(leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && !tau_truth_IsHadronicTau`
Fake τ	`(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && (leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && !tau_truth_IsHadronicTau`

Table A.5: Fake lepton re-classification

Re-Classification	Selection
$Fake\, l + \tau$	`(!(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) or !(leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7)) && !tau_truth_IsHadronicTau`
$Fake\, e$	`(!(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && (leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && tau_truth_IsHadronicTau && leps_flav_0_NOSYS = 0) or ((leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && !(leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && tau_truth_IsHadronicTau && leps_flav_1_NOSYS = 0) && tau_truth_IsHadronicTau`
$Fake\, \mu$	`(!(leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && (leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && tau_truth_IsHadronicTau && leps_flav_0_NOSYS = 1) or ((leps_IFFtype_0_NOSYS = 2 or leps_IFFtype_0_NOSYS = 4 or leps_IFFtype_0_NOSYS = 7) && !(leps_IFFtype_1_NOSYS = 2 or leps_IFFtype_1_NOSYS = 4 or leps_IFFtype_1_NOSYS = 7) && tau_truth_IsHadronicTau && leps_flav_1_NOSYS = 1) && tau_truth_IsHadronicTau`

Table A.6: Fake lepton re-classification

An "$IFF\_type = 2, 4, 7$" corresponds to prompt electron, prompt muon and lepton coming from τ decay. "$NOSYS$" means that the feature does not include the effect of systematics.

A.4 Feature Importance

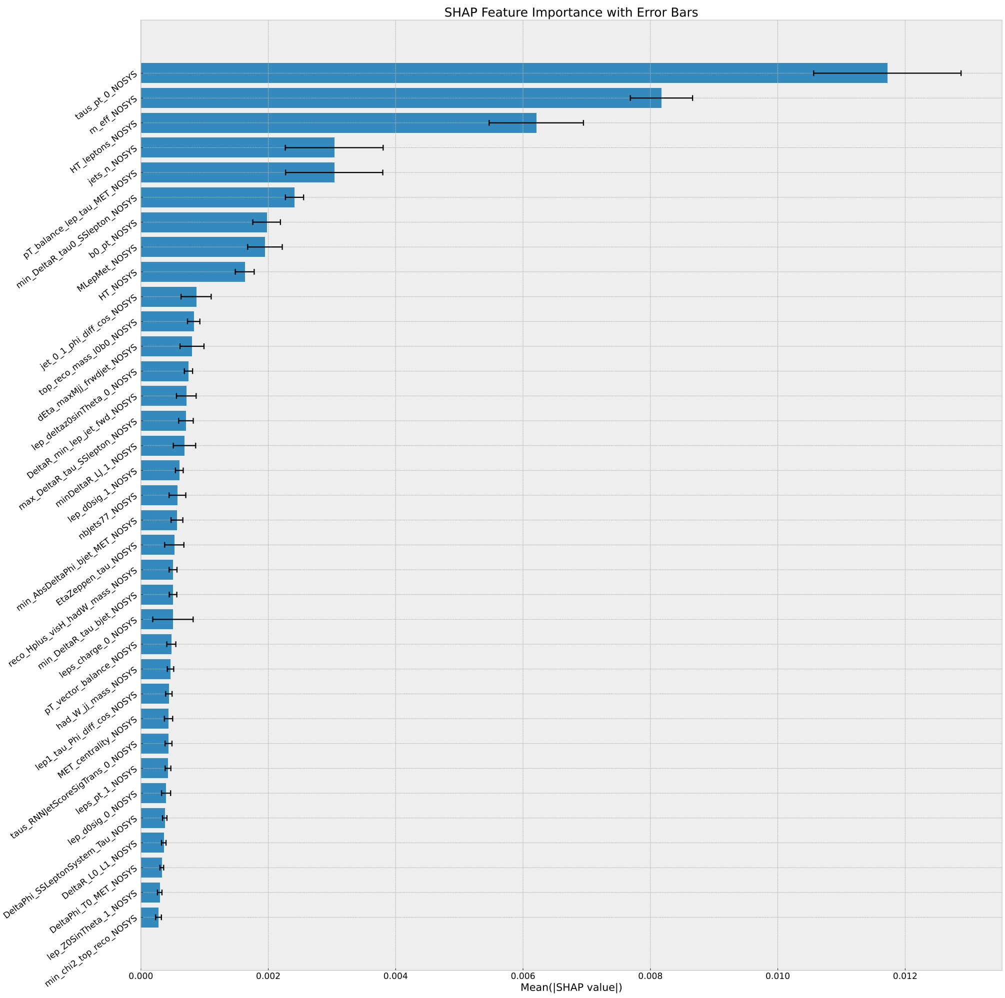


Figure A.1: Feature importance of all the features

A.5 Feature Correlation

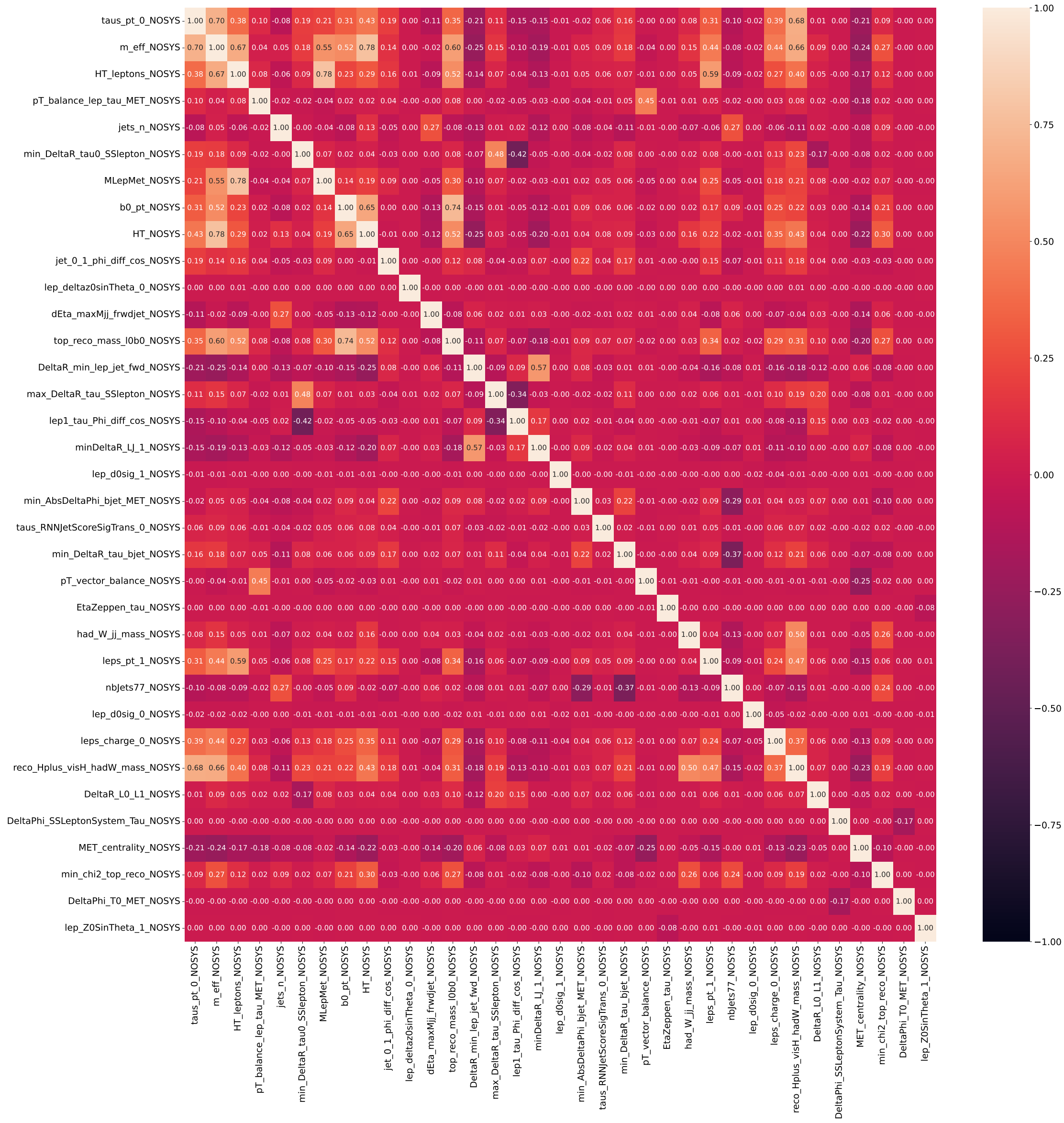
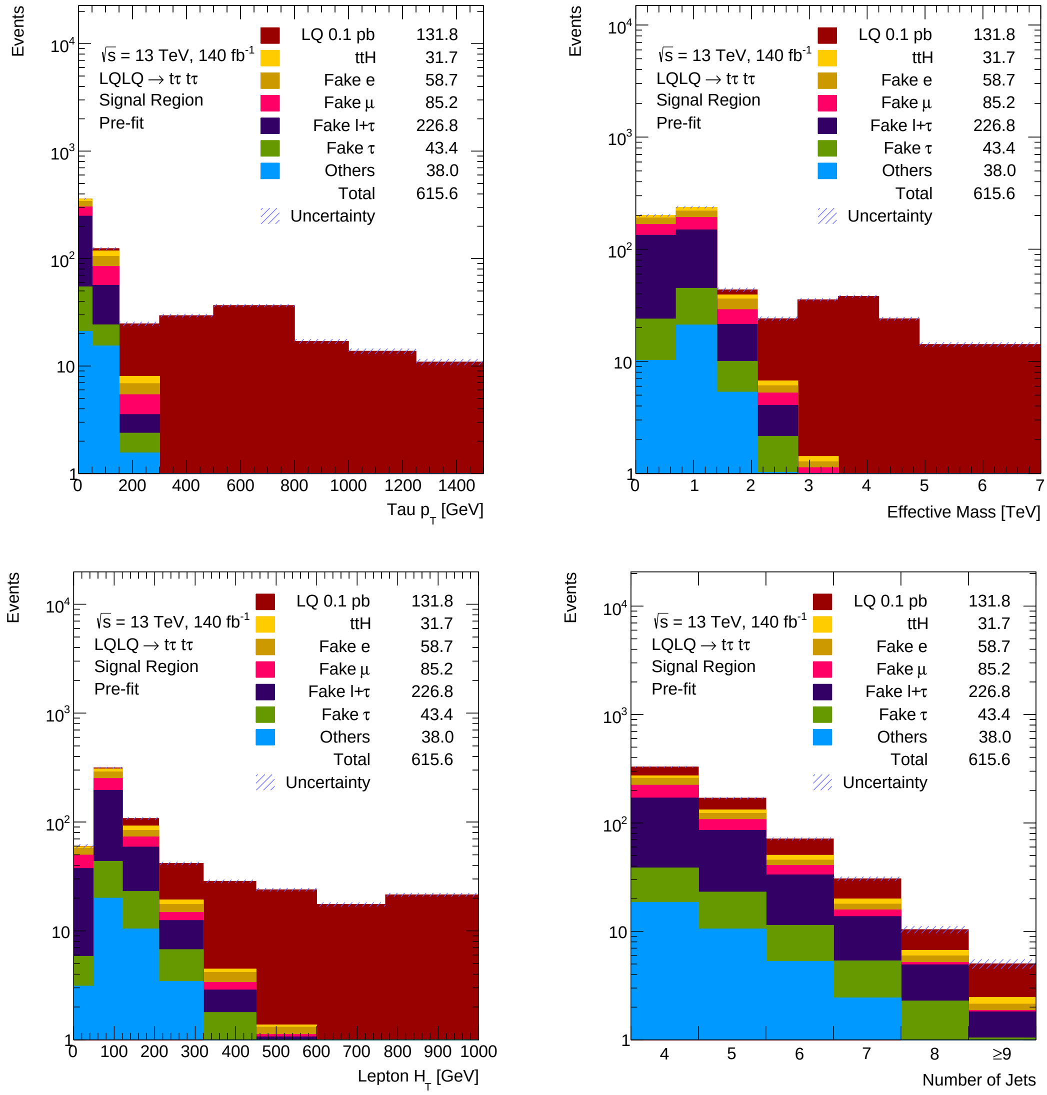
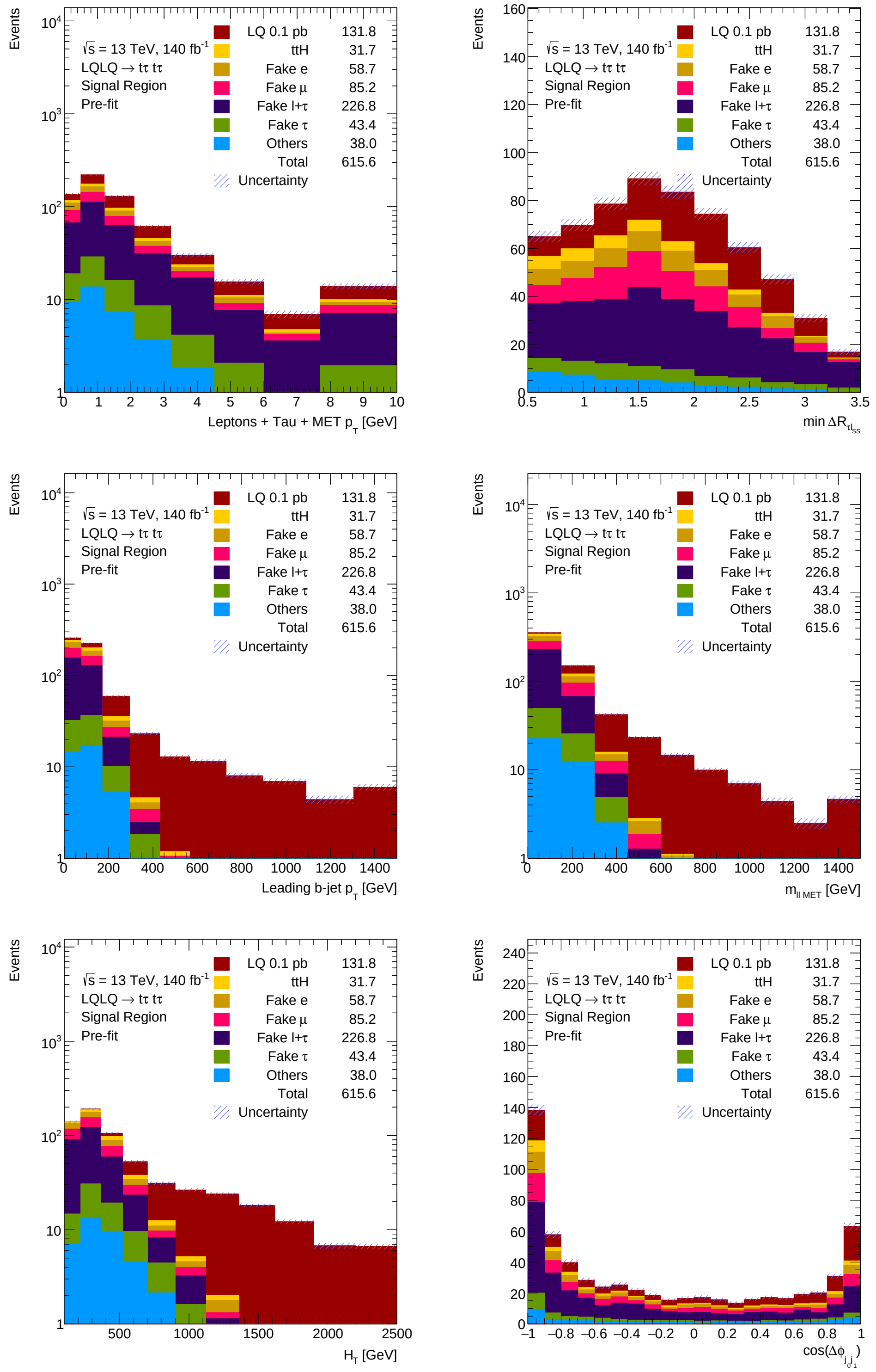


Figure A.2: Correlations of all the features

A.6 Distribution of Ten Most Important Features in the Signal Region




Events
√s = 13 TeV, 140 fb⁻¹
LQLQ → tτ tτ
Signal Region
Pre-fit
LQ 0.1 pb 131.8
ttH 31.7
Fake e 58.7
Fake μ 85.2
Fake l+τ 226.8
Fake τ 43.4
Others 38.0
Total 615.6
Uncertainty
Leptons + Tau + MET p_T [GeV]
min ΔR_τl_ss
Leading b-jet p_T [GeV]
m_ll MET [GeV]
H_T [GeV]
cos(Δϕ_j1j1)

Bibliography

[1] S. L. Glashow. "Partial-symmetries of weak interactions". In: *Nuclear Physics* 22.4 (1961), pp. 579–588. DOI: https://doi.org/10.1016/0029-5582(61)90469-2.

[2] S. Weinberg. "A Model of Leptons". In: *Phys. Rev. Lett.* 19 (21 Nov. 1967), pp. 1264–1266. DOI: 10.1103/PhysRevLett.19.1264.

[3] A. Salam. "Weak and Electromagnetic Interactions". In: *Conf. Proc. C* 680519 (1968), pp. 367–377. DOI: 10.1142/9789812795915_0034.

[4] G. Aad et al. "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC". In: *Physics Letters B* 716.1 (2012), pp. 1–29. DOI: https://doi.org/10.1016/j.physletb.2012.08.020.

[5] S. Chatrchyan et al. "Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC". In: *Physics Letters B* 716.1 (Sept. 2012), pp. 30–61. DOI: 10.1016/j.physletb.2012.08.021.

[6] G. Hiller and M. Schmaltz. "R_K and future $b \to s\ell\ell$ physics beyond the standard model opportunities". In: *Phys. Rev. D* 90 (5 Sept. 2014), p. 054014. DOI: 10.1103/PhysRevD.90.054014.

[7] U. Mahanta. "Neutrino masses and mixing angles from leptoquark interactions". In: *Phys. Rev. D* 62 (7 Sept. 2000), p. 073009. DOI: 10.1103/PhysRevD.62.073009.

[8] ATLAS Collaboration. *Analysis of $t\bar{t}H$ and $t\bar{t}W$ production in multilepton final states with the ATLAS detector*. Tech. rep. Geneva: CERN, 2019.

[9] B. Ali. *Search for the Standard Model Higgs boson produced in association with a top quark pair in multilepton final states at $\sqrt{s} = 13$ TeV with the ATLAS detector*. June 2021.

[10] A. Sopczak. *Searches for Leptoquarks with the ATLAS Detector*. 2021. arXiv: 2107.10094 [hep-ex].

[11] M. D. Schwartz. *Quantum Field Theory and the Standard Model*. Cambridge University Press, Mar. 2014.

[12] M. Thomson. *Modern particle physics*. New York: Cambridge University Press, Oct. 2013. DOI: 10.1017/CBO9781139525367.

[13] *The Standard Model of Particle Physics and Beyond*. https://opendata.atlas.cern/docs/documentation/introduction/SM_and_beyond. Accessed: 2025-08-22.

[14] J. C. Pati and A. Salam. "Lepton number as the fourth "color"". In: *Phys. Rev. D* 10 (1 July 1974), pp. 275–289. DOI: `10.1103/PhysRevD.10.275`.

[15] J. C. Pati and A. Salam. "Unified Lepton-Hadron Symmetry and a Gauge Theory of the Basic Interactions". In: *Phys. Rev. D* 8 (4 Aug. 1973), pp. 1240–1251. DOI: `10.1103/PhysRevD.8.1240`.

[16] I. Dorsner. "The Minimal SU(5) Unification Model". In: Nov. 2022, p. 075. DOI: `10.22323/1.406.0075`.

[17] D. Bečirević et al. "Scalar leptoquarks from grand unified theories to accommodate the B-physics anomalies". In: *Phys. Rev. D* 98 (5 Sept. 2018), p. 055003. DOI: `10.1103/PhysRevD.98.055003`.

[18] H. Georgi and S. L. Glashow. "Unity of All Elementary-Particle Forces". In: *Phys. Rev. Lett.* 32 (8 Feb. 1974), pp. 438–441. DOI: `10.1103/PhysRevLett.32.438`.

[19] I. Doršner et al. "Physics of leptoquarks in precision experiments and at particle colliders". In: *Physics Reports* 641 (June 2016), pp. 1–68. DOI: `10.1016/j.physrep.2016.06.001`.

[20] I. Doršner, S. Fajfer, and A. Lejlić. "Novel leptoquark pair production at LHC". In: *Journal of High Energy Physics* 2021.5 (May 2021). DOI: `10.1007/jhep05(2021)167`.

[21] E. Lopienska. "The CERN accelerator complex, layout in 2022". In: (2022). General Photo.

[22] L. Evans and P. Bryant. "LHC Machine". In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08001. DOI: `10.1088/1748-0221/3/08/S08001`.

[23] T. A. Collaboration et al. "The ATLAS Experiment at the CERN Large Hadron Collider". In: *Journal of Instrumentation* 3.08 (Aug. 2008), S08003. DOI: `10.1088/1748-0221/3/08/S08003`.

[24] R. M. Bianchi and ATLAS Collaboration. "ATLAS experiment schematic or layout illustration". General Photo. 2022.

[25] G. Cowan et al. "Asymptotic formulae for likelihood-based tests of new physics". In: *Eur. Phys. J. C* 71 (2011), p. 1554. DOI: `10.1140/epjc/s10052-011-1554-0`. arXiv: `1007.1727 [physics.data-an]`.

[26] A. L. Read. "Presentation of search results: the CL_s technique". In: *J. Phys. G* 28 (2002), p. 2693. DOI: `10.1088/0954-3899/28/10/313`.

[27] S. O. Arik and T. Pfister. *TabNet: Attentive Interpretable Tabular Learning*. 2020. arXiv: `1908.07442 [cs.LG]`.

[28] J. Alwall et al. "The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations". In: *JHEP* 07 (2014), p. 079. DOI: 10.1007/JHEP07(2014)079. arXiv: 1405.0301 [hep-ph].

[29] C. Bierlich et al. *A comprehensive guide to the physics and usage of PYTHIA 8.3*. 2022. arXiv: 2203.11601 [hep-ph].

[30] *TopCPToolkit*. https://topcptoolkit.docs.cern.ch/latest/. Accessed: 2025-09-08.

[31] T. Dado et al. *FastFrames*. Version 6.0.0. Apr. 2024. DOI: 10.5281/zenodo.16418658.

[32] M. Aly et al. *TRExFitter*. Version 1.4.0. Aug. 2025. DOI: 10.5281/zenodo.16684099.

[33] R. Brun and F. Rademakers. "ROOT – An object oriented data analysis framework". In: *Nucl. Instrum. Meth. A* 389.1 (1997), pp. 81–86. DOI: 10.1016/S0168-9002(97)00048-X.

[34] S. Mrenna and P. Skands. "Automated parton-shower variations in PYTHIA 8". In: *Phys. Rev. D* 94 (2016), p. 074005. DOI: 10.1103/PhysRevD.94.074005. arXiv: 1605.08352 [hep-ph].

[35] *TRUTH Derivations*. https://atlassoftwaredocs.web.cern.ch/analysis-software/AnalysisSWTutorial/mc_truth_derivation/. Accessed: 2025-09-10.

[36] *TauX_Gnt1makerAlg*. https://gitlab.cern.ch/atlas-phys/exot/lpx/exot-2022-34/taux_gnt1makeralg. Accessed: 2025-09-08.

[37] *TRUTH Derivations*. https://twiki.cern.ch/twiki/bin/view/AtlasProtected/DAODPhys. Accessed: 2025-09-10.

[38] G. Aad et al. "ATLAS data quality operations and performance for 2015–2018 data-taking". In: *Journal of Instrumentation* 15.04 (Apr. 2020), P04003–P04003. DOI: 10.1088/1748-0221/15/04/p04003.

[39] R. D. Ball et al. "Parton distributions for the LHC run II". In: *Journal of High Energy Physics* 2015.4 (Apr. 2015). DOI: 10.1007/jhep04(2015)040.

[40] R. D. Ball et al. "Parton distributions with LHC data". In: *Nuclear Physics B* 867.2 (Feb. 2013), pp. 244–289. DOI: 10.1016/j.nuclphysb.2012.10.003.

[41] I. Doršner, A. Lejlić, and S. Saad. "Asymmetric leptoquark pair production at LHC". In: *Journal of High Energy Physics* 2023.3 (Mar. 2023). DOI: 10.1007/jhep03(2023)025.

[42] M. Cacciari, G. P. Salam, and G. Soyez. "The anti-k_t jet clustering algorithm". In: *JHEP* 04 (2008), p. 063. DOI: 10.1088/1126-6708/2008/04/063. arXiv: 0802.1189 [hep-ph].

[43] G. Aad et al. “Tools for estimating fake/non-prompt lepton backgrounds with the ATLAS detector at the LHC”. In: *Journal of Instrumentation* 18.11 (Nov. 2023), T11004. DOI: 10.1088/1748-0221/18/11/t11004.

[44] A. Alkandari. *Determination of Fake Electrons, Muons and Taus in the tbH^+ Search for the 2lSS1tau Channel Using Template Fit Method - Summer Student Report*. Dec. 2023.